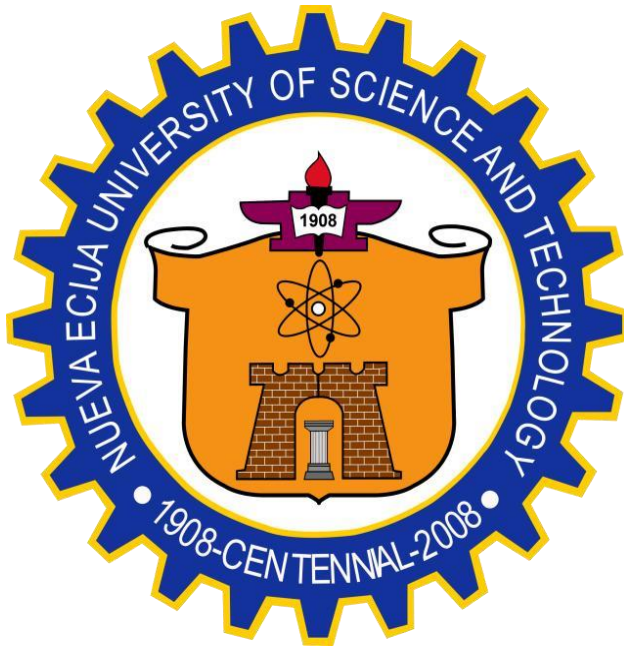


WEB-BASED RECORD MANAGEMENT AND DISPATCHING SYSTEM FOR TRUCKING SERVICES



A CAPSTONE AND RESEARCH PROJECT

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INTRODUCTION

Trucking services are essential to economic growth, ensuring the efficient movement of goods and supporting global trade. As the backbone of logistics, the trucking industry enables the timely delivery of raw materials, finished products, and essential supplies across vast distances. Industries such as manufacturing, retail, construction, and agriculture depend on freight transportation to maintain supply chain efficiency and meet consumer demands (Park et al., 2024). Beyond its logistical role, trucking is also a business that relies on accurate invoicing to compute income. Each trip generates an invoice detailing transportation costs, fuel expenses, toll fees, and other charges, which are essential for tracking revenue and managing operational expenses (Milambiling et al, 2023). However, despite its critical role in commerce, the trucking industry faces persistent operational challenges that affect its financial stability.

The Philippine Trucking Industry continues to face significant challenges due to its heavy reliance on outdated manual processes, particularly in financial and administrative operations. Many trucking companies still manage their invoicing through paper-based records or excel spreadsheets, requiring manual input for every trip Sanchez (2024) . This approach not only consumes a considerable amount of time but also increases the likelihood of errors, making invoice management inefficient and unreliable. Personnel responsible for invoicing often encounter issues such as disorganized records, redundant entries, and miscalculations, resulting in confusion and operational delays (Shankar, 2023) . Without proper system, essential tasks such as tracking payments, verifying trip details, and maintaining accurate financial records become overwhelming and difficult to manage. These inefficiencies extend beyond administrative burdens, as they contribute to delays

in payment processing, disrupt cash flow, and weaken the company's overall financial stability (Mioten, 2023). As a result, many trucking services struggle to optimize their financial operations, which emphasized the urgent need for modernization and digital transformation in the industry.

To address these challenges, the researchers propose to develop a Web-Based Record Management and Dispatching System for Trucking Services. This system aims to replace outdated manual processes with a semi-automated platform that simplifies invoice management, expense tracking, and trip coordination. By organizing financial records such as expenses, and trip details, it will enhance accuracy, reduce errors, and eliminate redundant data entry. Additionally, Showing which trucks are available makes it quicker and easier to choose which vehicles and drivers can be dispatched.

The system will be introduced in a selected trucking services operating in Muralla Street, Meycauayan, Bulacan with a focus on improving financial management, dispatch coordination, and record accuracy. Through semi-automation, local trucking businesses will benefit from more reliable invoicing, simplified expense monitoring, and optimized resource allocation.

By integrating this digital solution into trucking operations, the system will help businesses minimize errors, enhance service reliability, and support long-term industry growth and sustainability.

Review of Related Literature

Traditional Record Management

Traditional record management in the trucking industry faces significant challenges that hinder both efficiency and compliance. Manual processes involve extensive paperwork and administrative effort, often resulting in delays and errors when managing driver

logs, vehicle maintenance records, and dispatch schedules (Basopo, 2023).

Furthermore, Yang et al. (2023) highlight that storage limitations in traditional systems pose major issues, as physical records occupy valuable space and are susceptible to misplacement, making it difficult to retrieve critical data when needed. Paper records are also vulnerable to damage or loss due to disasters or theft, complicating audits. Inconsistent filing, illegible handwriting, and missing documents further exacerbate these problems, causing confusion and negatively affecting fleet scheduling and regulatory compliance.

Technology in Logistics and Dispatching

Ayo et al. (2023) examines the factors influencing and hindering technology adoption in logistics management. This study emphasizes the importance of integrating new technologies to enhance process management, monitoring, and overall efficiency in logistics operations.

The study recommends increased investment in technology, effective integration of new and existing systems, establishment of industry standards, and enhanced employee participation to overcome barriers to technological adoption in the logistics sector.

Digital Era of Trucking Industry

The trucking industry is gradually embracing digital transformation to enhance efficiency and adaptability. Historically dependent on manual processes and paper-based documentation, the sector is now integrating digital tools to improve communication, tracking, and decision-making (Remez 2024). Automation technologies, such as route optimization software is being adopted to streamline operations, reduce fuel consumption, and improve delivery times. (Turvo 2024).

Transformation of Paper-based to Automation

Despite the potential benefits, the digital transformation of the trucking industry faces significant obstacles. Resistance to change among some of the traditional operators slows down progress. Additionally, the transition to digital systems comes with challenges such as cybersecurity risks, data management complexities, and the need for extensive employee training. Rather than ensuring seamless efficiency, these issues can lead to operational disruptions and increased expenses (Paula 2023).

Website Usability

Nomensa (2024) highlights that website usability is crucial for user satisfaction, engagement, and retention, in which it was stated that well-designed website should have clear navigation, an intuitive user interface, structured content, and engaging visuals to enhance accessibility and user experience. Additionally, fast-loading pages contribute to a smoother interaction, reducing frustration. Moreover, usability testing is essential in identifying and addressing common issues, such as inconsistent layouts and complex menus, which can negatively impact user experience.

Application of Web-Based Systems in Trucking Industry

According to Mohamed et al. (2024), web-based databases help businesses grow by giving them important information for better decision-making. They make it easier for companies to track their financial performance and allow different teams to work together more smoothly. Using databases also helps reduce risks, manage resources better, and improve efficiency. This keeps businesses competitive, helps avoid losses, and increases profits.

Cybersecurity Risks

As businesses increasingly rely on digital systems, they are exposed to growing cybersecurity risks, such as data breaches,

unauthorized access, and various forms of cyberattacks. These threats pose significant challenges to securing sensitive information and maintaining operational integrity. To address these challenges, solutions such as advanced authentication methods and continuous user verification play a critical role in mitigating these risks. Implementing strong authentication protocols ensures that only authorized users gain access to sensitive systems, while continuous verification helps to reduce the chances of unauthorized access, thereby strengthening overall security (Admin, 2023).

Environmental Impact of Paper-based to Digital

Switching from paper to digital systems greatly helps the environment by reducing paper use. Harikrishan et al. (2025) explain that traditional paper-based methods use a lot of resources, like paper, ink, and energy for printing and distribution. This leads to more deforestation and higher carbon emissions. Moving to digital reduces waste and lowers the environmental impact of paper production and disposal.

Importance of Expense Tracking

Tracking expenses is important in managing a business's finances effectively in order not to lose profit. It allows companies to monitor cash flow, control costs, and create accurate budgets. By keeping detailed records of expenses, businesses can identify areas where they can reduce unnecessary spending and allocate resources more efficiently. Proper expense tracking also ensures compliance with financial regulations, and helps in financial forecasting. Additionally, having a clear record of expenses provides insights that can improve profitability, and support long-term business growth while preventing financial losses (Bhuvanapriya et al., 2024).

Cost Challenges

One of the challenges in adopting a trucking management system is the substantial initial investment required. The implementation of such a system involves costs for software acquisition. For many trucking companies, particularly smaller ones, this upfront financial outlay can be a significant barrier, potentially hindering their ability to adopt such technology (Chaves 2023).

Fleet Management System

Sanchez (2024) highlights that fleet management software can solve issues like inefficiency and regulatory challenges in the trucking industry. Key features include route planning, vehicle maintenance, driver and fuel management, compliance tools, and data analytics. The study shows digitalization improves efficiency, reduces costs, boosts safety, and supports better decisions. It also notes adoption hurdles like high initial costs, tech limitations, and resistance to change, and suggests solutions like phased rollouts, training, partnerships, and cybersecurity focus.

Proof of Delivery

Proof of Delivery software enhances delivery confirmation and job management by replacing traditional paper-based systems with a mobile application. The software captures real-time data such as delivery photos, and timestamp. This system enhances delivery accuracy. Moreover, it offers businesses real-time visibility into their delivery operations, leading to better tracking (Endres 2024).

Enhanced Invoice Processing

Oana et al. (2025) proposed a study that enhance invoice processing automation by integrating development and operation (DevOps) methodologies with machine learning. This approach seeks to reduce manual effort, improve efficiency, minimize errors, and

optimize resource utilization. The study also looks at how using DevOps and machine learning together can make financial processes smarter and more flexible. This means businesses can quickly adjust to changes and make better decisions. At the same time, it helps reduce the environmental impact by using fewer resources and producing less waste.

Payroll Invoice

Ahmed et al. (2023) pointed out that manual payroll processing is slow, prone to errors, and causes delays in generating reports. This approach often leads to mistakes in salary calculations, incorrect allowances and deductions, and general inefficiencies in handling payroll data. To solve these issues, they created a web-based payroll management system that improves salary calculations, allowances, deductions, and recordkeeping. The system received a high score of 87.8% on the System Usability Scale (SUS), showing strong user satisfaction. This solution effectively fixes the problems of manual payroll systems, making the process faster and more reliable, especially for organizations in developing regions.

Web-based Document Management System

Aládé (2023) developed the Web-based Document Management System (WBEDMS) to improve document handling and organizational efficiency. It features version control, ensuring that changes to documents are tracked, and audit trails to monitor user activity. The system includes advanced search functions to quickly locate documents and role-based access controls to ensure security and proper authorization.

Automated Data Management

Booc et al. (2022) developed the "Z Billing and Invoice Application" for Kwik Fit Tyre Depot to semi-automate and improve the management of daily transactions. The system minimizes

paperwork and enhances data organization through a database management system, allowing secure storage and easy retrieval of customer billing records for backup and restoration. It also features reporting modules—such as billing summaries—that support effective monitoring of activities from daily to annual periods.

Dispatching System

Laurinte (2024) developed a Transportation Management System for the Fatima Airport Transportation Cooperative to simplify the dispatching process. The system includes a visual dashboard that allows dispatchers to easily see which vehicles are available or unavailable. It replaces manual tasks by automating scheduling, driver management, and data handling. By centralizing data, it securely stores driver and dispatcher records, making backups and restorations easier. The system also has reporting features, like scheduling logs, to help track dispatching activities. This leads to better efficiency, fewer errors, improved communication, and more informed decision-making with real-time data and performance tracking.

SYNTHESIS

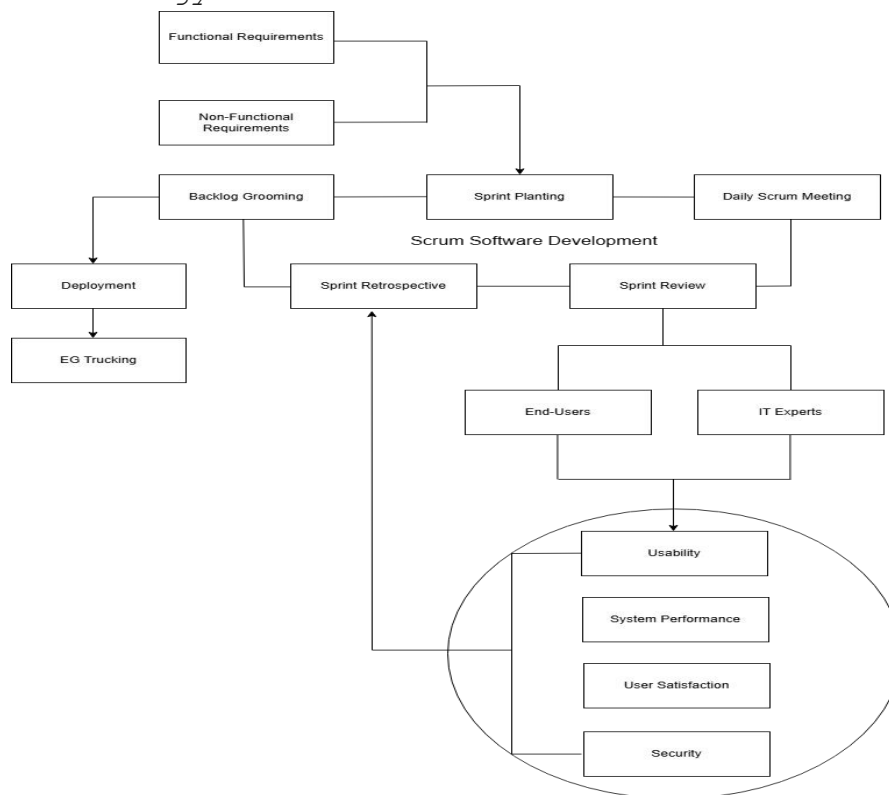
The research reviewed provides a solid foundation for developing a Web-Based Record Management and Dispatching System for Trucking Services. By looking at existing studies, the researchers were able to identify key similarities and differences, which helped guide the design of the proposed system. Previous studies shows the challenges trucking businesses face with manual invoicing and record-keeping, as well as the difficulties of ensuring compliance and accuracy. Many of these studies also point out how digital solutions can address these issues, improving efficiency and reducing the risks tied to traditional paper-based methods.

Earlier systems have made use of automation and data integration tools to streamline processes and improve data management. These technologies aim to create more reliable, efficient, and secure systems. The goal of this research is to build a web-based platform that incorporates these technologies to offer a semi-automated and efficient solution for record management and dispatching. By doing so, the system will reduce paperwork, minimize errors, and improve overall operational performance for trucking companies.

CONCEPTUAL FRAMEWORK

Figure 1

Agile Methodology



This study uses the Agile methodology to develop a Web-Based Record Management and Dispatching System for Trucking Services. This approach allows continuous improvements based on user feedback and changing business needs.

The development begins with planning, where the team outlines key features. The system will handle dispatching, fleet tracking, expense monitoring, billing and invoicing, and route optimization. It will also track toll costs, generate semiautomated invoices, and find the best routes to save time and fuel. Additionally, it ensures security, efficiency, and ease of use.

To maintain progress, the team holds regular meetings to discuss updates and address challenges. After each development cycle, a review is conducted with trucking owners to evaluate expense tracking, invoice, route optimization, and overall system performance.

Following the review, a reflection session helps the team identify successful aspects and areas for improvement. Continuous updates ensure the system evolves, incorporating enhancements like better expense tracking, accurate invoicing, and smarter route planning.

Finally, the system is deployed and becomes fully operational. Trucking companies can then use it to manage fleets, track expenses, salary invoice, and optimize routes, improving overall efficiency.

By following an iterative and adaptive development process, this project ensures continuous improvements, regular feedback, and a reliable system for modern trucking operations.

Statement of the Problem

This study aims to develop and evaluate a Web-Based Record Management and Dispatching System for Trucking Services to automate financial transactions, optimize dispatch coordination, and enhance administrative efficiency.

Specifically, it seeks to address the following:

1. How may the current system be described in terms of the following constructs?

- 1.1. Security of files and records;
- 1.2. Management of data;
- 1.3. Reliability of output;
- 1.4. Efficiency of processes; and
- 1.5. Compliance with regulations and standards?

2. How may the respondents develop the application based on the following software development phases?

- 2.1. Planning and Requirements Gathering;
- 2.2. Design and Prototyping;
- 2.3. Development and Implementation; and
- 2.4. Testing and Deployment?

3. How may the elected IT experts assess the proposed system according to the following software characteristics:

- 3.1. Functional suitability;
- 3.2. Performance efficiency;
- 3.3. Compatibility; 3.4. Usability; and
- 3.5. Security?

4. How may the end-users evaluate the developed system based on the System Usability Scale (SUS) in terms of the following aspects:

- 4.1. Ease of use;
- 4.2. Efficiency;
- 4.3. Learnability; and
- 4.4. User Satisfaction?

Scope and Delimitations

This study focuses on developing a Web-Based Record Management and Dispatching System for Trucking Services to improve financial and administrative operations. The system is designed to semi-automate salary invoicing, expense tracking, route optimization, and dispatch coordination, replacing outdated manual processes that contribute to inefficiencies and errors. By streamlining these tasks, it enhances record accuracy, minimizes redundant data entry, and improves overall operational efficiency.

The system includes semi-automated invoicing to compute transportation costs and expenses, salary invoicing for drivers and helpers, dispatch coordination to efficiently assign drivers and helpers, and financial record management to securely store and access trip and payment data. Additionally, it offers route optimization to ensure the most efficient paths are taken, saving time and reducing costs. Applicants like drivers and helpers can apply directly through the site by submitting their requirements. To safeguard sensitive financial information, the system integrates user access controls and security measures, ensuring the privacy and integrity of all records.

This study aims to address key challenges in trucking services, such as inefficient manual record-keeping, delays in dispatch coordination, and the lack of a centralized financial tracking system. The system is developed specifically for a selected trucking company to enhance operational efficiency within its business processes. The project follows the Agile methodology, ensuring continuous testing, feedback, and improvements throughout development. Evaluation focuses on effectiveness, usability, and security, with insights gathered from trucking company owners, dispatch personnel, and IT experts. The research is conducted from January to December 2025, aligning with the project timeline established by the researchers.

This study is delimited to a selected trucking services in Muralla Street, Meycauayan Bulacan and does not extend to other logistics sectors, such as shipping or air freight. It primarily focuses on record-keeping and dispatching. Furthermore, as a webbased system, it requires an active internet connection for full functionality.

Significance of the Study

The results of this study will help the following:

Trucking Business Owner - The system will help trucking business owners by automating financial records and dispatch operations. It will make it easier to track expenses, create invoices, and manage trips, helping them run their business more efficiently.

Dispatch Personnel - The system will help dispatchers organize trip schedules and assign drivers more effectively. It will reduce errors and miscommunication, making the dispatch process smoother.

Drivers and Helpers - The system will provide clear trip details and schedules, ensuring that drivers and helpers receive the correct information. This will help them avoid confusion and delays in deliveries.

IT Specialists and Developers - This study will serve as a reference for software developers working on logistics and financial management systems. It will show how the Agile Scrum Model can be used to develop an efficient and reliable web-based system.

Future Researchers - This study can serve as a guide for future research related to logistics, transportation, and financial automation. Future researchers can build on this system and explore new technologies like artificial intelligence or blockchain for trucking services.

The Local Trucking Industry - This system will help modernize trucking services in the Philippines by reducing manual work, improving financial accuracy, and making dispatching more efficient.

Definition of Terms

The following terminologies are defined according to their dictionary meaning and their usage in the system for better understanding.

Authentication - The process of verifying the identity of a user, system, or device to ensure secure access to a platform or service (Dela 2023). In the trucking industry, authentication is crucial for protecting sensitive data in web-based record management and dispatch systems. It ensures that only authorized personnel, such as dispatchers, drivers, and administrators, can access and manage financial records, trip details, and other critical business information.

Cash Flow - The total amount of money being transferred in and out of a business, from revenues and expenses (Hayes 2023). In trucking industry, it is primarily through freight revenues, fuel costs, maintenance expenses, driver wages, and other operational costs. Positive cash flow ensures a trucking company can cover expenses and grow, while negative cash flow may cause financial strain and disrupt operations.

Dispatch System - is a structured process used to assign and coordinate the movement of trucks and drivers, ensuring efficient scheduling, real-time tracking, and timely deliveries. According to Choudhury (2022), an efficient dispatch system enhances productivity by optimizing truck allocation, reducing idle time,

improving delivery schedules, and facilitating accurate and timely communication between drivers and dispatchers.

Payroll Invoice - document or statement that details the payment made to employees for their work over a specific period. It includes a breakdown of earnings, deductions, and net pay.

Profit- is a key measure of financial success and sustainability, calculated as the difference between income and costs (Jebunnesa 2023). In the trucking industry, it refers to the financial gain a company earns after subtracting all operating expenses, including fuel, maintenance, driver wages, insurance, and toll fees, from its total revenue.

Semi-Automation - refers to a system or process that combines both manual and automated elements. In semi-automation, certain tasks are automated, while others still require human intervention. This hybrid approach allows for increased efficiency and accuracy compared to fully manual processes, but it doesn't fully eliminate the need for human input or oversight.

Trucking Services - refer to businesses or operations that provide transportation of goods using trucks. These services typically involve the moving of materials from one location to another, either locally or across long distances.

Chapter II

METHODOLOGY

This section presents a systematic approach to efficiently developing and implementing the study. It includes key components such as the research design, locale, target respondents, and data analysis methods.

Research Design

The study will employ a descriptive-developmental research design while developing the Web-Based Record Management and Dispatching System for Trucking Services using an Agile Software Development Approach, specifically the Scrum framework. The study will examine current trucking management issues and evaluate the effectiveness of the proposed solution.

The developmental aspect will focus on designing, developing, and refining the system, while the descriptive part will look at the current manual processes in place. Agile methods, especially Scrum, are known for their adaptability, quick prototyping, and continuous feedback loops. These characteristics will be essential for adjusting to changing requirements and ensuring the system meets user needs and organizational goals. Regular review sessions will allow stakeholders to provide feedback, leading to quick adjustments and better development efficiency. Surveys and system performance assessments will be used as quantitative data collection methods throughout the study.

Locale

The study will be conducted at Muralla Street, Meycauayan, Bulacan, with an emphasis on the selected trucking services, Eugene Galvez Trucking Services. This was selected because it operates multiple trucks, and currently lacks a formal system for managing operational data such as trip expenses, fuel, maintenance, driver availability, and payroll.

Figure 2

Map of Eugene Galves Trucking Services



Figure 3 shows the map of Eugene Galves Trucking Services, located at Muralla Street Meycauayan, Bulacan.

Figure 3

Eugene Galves Trucking Services Operational Site



Figure 3 shows the operational site of Eugene Galves Trucking Services, located at Muralla Street Meycauayan, Bulacan.

Respondents

The respondents to the study will be the key stakeholders in the trucking industry who will be chosen using purposive sampling

to ensure that relevant insights were collected. The participants were divided into the following groups:

Owners of trucking business will evaluate the system's financial and operational benefits. Since dispatching is frequently handled by business owners, their opinions will be very important when evaluating dispatch effectiveness. In addition, drivers and helpers will offer insights on system usability in terms of route optimization, and trip assignments. Lastly, elected IT Experts and faculty members will determine the technical and functional quality of the system.

Table 1

Distribution of the Respondents

Respondents	Frequency	Percentage
Owner	2	5.56%
Driver	10	27.78%
Helper	14	38.89%
IT Experts And Faculty Members	10	27.78%
TOTAL	36	100%

Table 1 shows the distribution of the respondents with a frequency of 31.

Sample and Sampling Procedure

Purposive sampling, a non-probability sample strategy, will be used in the study. Participants will be purposely chosen based on certain traits relevant to the research objectives.

The quantity of owner, drivers, and helpers of Eugene Galves

Trucking Services located in Muralla Street, Meycauayan, Bulacan will be used to calculate the sample size. The following are among the selection criteria for respondents: (a) engagement in trucking activities in the research area. (b) familiarity with manual dispatching and record-keeping procedures. (c) willingness to take part in system evaluation-related questionnaires.

Additionally, the researchers will establish the following criteria for selected IT experts and faculty members: (a) must have completed a degree in IT and (b) must have experience in the information technology industry.

Research Instrument

The tool for this study is utilized to collect, measure, and assess data relevant to the research objectives. A survey questionnaire will use to evaluate the common challenges encountered in the manual dispatch and record management system used by trucking services. The questionnaire focuses on five key categories: security of files and records, management of data, reliability of output, efficiency of processes, and compliance with regulations and standards.

To assess the quality and technical aspects of the developed system based on software quality standards, the researchers will utilize evaluation questionnaires targeted at IT consultants and end-users. These questionnaires consist of closed-ended questions designed to gather quantitative feedback. The IT consultants' questionnaire is divided into five components based on software quality characteristics: functional suitability, performance efficiency, compatibility, usability, and security. The end-user questionnaire follows the System Usability Scale (SUS) and is

structured into four (4) components: ease of use, efficiency, learnability, and user satisfaction.

Data Gathering

The following phases formed the basis for the procedures of this study.

Phase I: Assessment of the Manual System

The researchers will evaluate the current manual dispatch and record management system used by trucking services to identify problems in invoice tracking, dispatching, and recordkeeping. Data will be gathered through survey questionnaires with administrative staff. The collected information will help define system requirements, ensuring the web-based solution effectively addresses these challenges.

Phase II: System Development and Initial Testing

Using the gathered data, the researchers will design and develop the web-based record management and dispatch system. This phase includes creating the system architecture, database structure, and user interface, integrating key features that will address the challenges encountered in manual record-keeping. Initial testing will be conducted to assess system functionality, with adjustments made based on identified issues and recommendations from trucking service personnel.

Phase III: System Pre-Testing and Evaluation

Before full deployment, the developed system will undergo pre-testing with trucking company to evaluate its usability, reliability, and performance. A review will be conducted to identify any flaws in functionality, security, or efficiency. The results will be analyzed, and necessary refinements will be made

based on feedback from trucking service personnel involved in the testing process.

Phase IV: Final Testing and Data Collection

In the final phase, the system will be tested in a realworld trucking service environment. The researchers will seek authorization from trucking service administrators and personnel before distributing evaluation forms and collecting feedback. The system will be assessed based on software quality standards, including functionality, efficiency, and user satisfaction. The researchers will remain open to additional suggestions from respondents to further enhance the system before its full implementation.

Data Analysis Technique

While gathering the needed data, the researchers will employ the following data analysis techniques:

1. To address the first statement of the problem regarding the security of files and records, data management, output reliability, process efficiency, and adherence to regulations and standards, frequency and weighted mean will be utilized to analyze the dataset and interpret the evaluation results provided by the group of respondents on the current tracking system.

The scoring rubric in Table 2 will be used to interpret the results correctly.

Table 2

Scoring Rubric for Level of Agreement

Range Description	Remarks	Verbal
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3.26 - 4.00

Strongly Agree The range of 3.26 to 4.00 is considered "Strongly Agree" in the quantitative description. This means the respondent has a high level of agreement with the statement or question. It indicates that they strongly supports the presented statement.

Range	Remarks	Verbal Description
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2.51 - 3.25	Agree	The range of 2.51 to 3.25 is classified as "Agree," indicating that the respondent has a favorable view of the statement but not as strong as those in the "Strongly Agree" category.
1.76 - 2.50	Disagree	The range of 1.76 to 2.50 is classified as "Disagree," which indicates that the respondent has an unfavorable view of the statement or question being
1.00 - 1.75	Strongly Disagree	The range of 1.00 to 1.75 is classified as "Strongly Disagree," which indicates that the respondent has a high negative or unfavorable view of the statement or

Table 2 presents the scoring rubric for the level of agreement, outlining the range, and verbal descriptions used in the four-point scale.

2. For the second statement of the problem, the researchers will outline the activities in each phase of the software development lifecycle model, offering relevant insights into the processes to be carried out.

Planning and Requirements Gathering. In this phase, the researchers will begin at identifying the specific problems the system aims to solve, such as inefficiencies in managing records. They will gather input from stakeholders, including trucking managers to determine the core features and requirements of the system. Surveys will be used to understand user needs and define functional and non-functional requirements.

Design and Prototyping. Researchers will then design the system architecture, user interface, and workflow diagrams. Mockups will be created to visualize how the system will look and function. At this stage, they will use prototyping tools to present a working model of the system to stakeholders, allowing for early feedback and refinement before full development.

Development and Implementation. Based on the approved design, the actual coding of the application will take place. Researchers will use a programming language and framework suited for web-based systems. During this phase, key modules such as user login, record tracking, and dispatch scheduling are developed. The system will also be implemented incrementally to allow continuous integration and review.

Testing and Deployment. Finally, the system will be tested for bugs, security issues, and usability concerns. This includes unit testing, integration testing, and user acceptance testing. Once verified, the system will be deployed in a live environment for actual use by the trucking company. Researchers will also provide documentation and training to users for smooth transition and adoption.

3. To address the third statement of the problem regarding the IT expert's evaluation of the system's qualifications, the

researchers will apply the following software characteristic guidelines to gain a better understanding of the enhanced evaluation results.

Table 3 presents the scoring rubric for Functional Suitability.

Table 3

Scoring Rubric for Functional Stability

Range	Remarks	Verbal Description
3.26 - 4.00	Highly Suitable	The system meets or exceeds all functional requirements with a high level of accuracy, completeness, and efficiency.
2.51 - 3.25	Suitable	The system meets most of the functional.
1.76 - 2.50	Unsuitable	The system fulfills the requirements with acceptable levels of accuracy, completeness, and efficiency.
		However, some functional requirements are met, but with limited accuracy, completeness, and efficiency.
Range	Remarks	Verbal Description
1.0 - 1.75	Highly Unsuitable and lacks	The system does not meet most or all functional requirements accuracy, completeness, and efficiency.

As shown in Table 3 above, the scoring rubric will use to assess the system's functionality under specific conditions.

Table 4 depicts the scoring rubric for Performance Efficiency.

Table 4

Scoring Rubric for Performance Efficiency

Range	Remarks	Verbal Description
3.26 - 4.00	Excellent	The system's performance is exceptional, meeting or exceeding all efficiency requirements. Users experience no delays or errors, and the system performs consistently well under all operating

Range	Remarks	Verbal Description
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2.51 - 3.25	Good	The system's performance is above average and meets most efficiency requirements. Users experience minimal delays and errors, and the system performs well under normal operating
1.76 - 2.50	Fair	The system's performance is average, but there is room for improvement. Users may experience occasional delays or errors, but the system functions adequately overall.
1.00 - 1.75	Poor	The system's performance could improve and meet the minimum requirements for efficient operation. Users may experience delays, errors, and other issues negatively affecting productivity.

Depicted in Table 4 is the rubric that will be used to determine the capacity of a system to carry out its intended function within a predetermined time.

Table 5 shows the scoring rubric for Compatibility.

Table 5*Scoring Rubric for Compatibility*

Range	Remarks	Verbal Description
3.26 - 4.00	Highly compatible	
2.51 - 3.25	Compatible	The system or software is fully compatible with other hardware or software and can seamlessly integrate.
1.76 - 2.50	Somewhat compatible	The system or software can work with hardware or software with minor adjustments
1.00 - 1.75	Not compatible	The system or software can work with other hardware or software with significant adjustments or configurations. The system or software cannot work with other hardware or software and needs a complete overhaul or replacement.

Shown in Table 5 above is the rubric that will be used to measure the ability of a system to share resources and an environment with another system.

Table 6 presents the scoring rubric for Usability.

Table 6*Scoring Rubric for Usability*

Range	Remarks	Verbal Description
3.26 - 4.00	Excellent	easily accomplish their tasks. The system is well-designed, intuitive, and efficient. Users rarely encounter errors or obstacles and are highly satisfied with the system.
2.51 - 3.25	Good	The system is generally easy to use, but there may be some minor usability issues or areas where improvement is needed. Users may encounter occasional obstacles or errors, but these do not significantly impede their ability to complete tasks. Users are generally satisfied with the system.
1.76 - 2.50	Fair	The system has significant usability issues that make it frustrating for users to accomplish tasks.
The system is easy to use and navigate, and users can		
Range	Remarks	Verbal Description

		be are the
		The system may need to better designed. Users generally dissatisfied with system.
1.00 - 1.75	Poor	The system is difficult or impossible to use, and users cannot accomplish their tasks. The system may be unusable due to significant errors or obstacles. The system is poorly designed or requires significant effort to use. Users are highly dissatisfied with the system.

Presented in Table 6 above is the rubric that will used to analyze the usability of a system by being able to communicate and transfer data via the user interface between a user and a system.

4. To answer the fourth statement of the problem, the end users' evaluation of the system's quality based on the System Usability Scale (SUS), the researchers will utilize the following software characteristics guidelines to improve evaluation outcomes.

Table 7*Scoring Rubric for Ease of Use*

Range	Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	Users find the system highly intuitive and effortless to navigate. They quickly grasp its terminology and can easily adapt to new features as they are introduced.
2.52 - 3.25	Agree	Users feel comfortable using the system, demonstrating a good understanding of its interface and functions. Still, navigation remains smooth.
1.76 - 2.50	Disagree	Users experience some difficulty in navigating the system and require additional effort to understand its features and terminology.
1.00 - 1.75	Strongly Disagree	Users find the system challenging to use, struggling to navigate its interface and comprehend its functions. Significant assistance may be required for effective usage.

Shown in Table 7 above is the rubric that will use to evaluate the system's ease of use. This scale will measure how effortlessly users can navigate and interact with the system, including their ability to understand its interface, access features, and complete tasks efficiently.

In table 8, the scoring rubric for Learnability is shown.

Table 8

Scoring Rubric for Learnability

Range	Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	understanding its features and functions with minimal guidance. New tasks feel intuitive and require little effort to master.
2.52 - 3.25	Agree	Users adapt to the system at a reasonable pace, requiring some initial learning but gradually becoming comfortable with its features and functions.
1.76 - 2.50	Disagree	Users experience moderate difficulty in learning how to use the system, needing additional time or assistance to understand its features and processes.
Users find the system easy to learn, quickly		
Range	Remarks	Verbal Description

1.00 - 1.75	Strongly Disagree	Users struggle to learn the system, finding it complex and unintuitive. Significant training and guidance are necessary to perform basic tasks.
-------------	-------------------	---

As shown in Table 8, the rubric for a system's learnability will be used to assess how easily users can understand and adapt to its features and functions. This evaluation determines the system's intuitiveness, the time required for users to become proficient, and the level of assistance needed for effective use. In table 9, the scoring rubric for User Satisfaction is depicted.

Table 9
Scoring Rubric for User Satisfaction

Range	Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	Users are highly satisfied with the system, finding it efficient, reliable, and enjoyable to use. It meets their expectations and enhances their overall

Range	Remarks	Verbal Description
-------	---------	--------------------

2.52 - 3.25 Agree

Users are generally satisfied with the system, though there may be minor areas for improvement. It meets most of their needs and functions effectively.

1.76 - 2.50 Disagree

Users have a moderate level of satisfaction, encountering some challenges or limitations that affect their overall experience.

1.00 - 1.75 Strongly
Disagree

Users adapt to the system at a reasonable pace, requiring some initial learning but gradually becoming comfortable with its features and functions.

As depicted in Table 9, the rubric for assessing user satisfaction will be used to evaluate how well the system meets user expectations, provides a positive experience, and fulfills their needs efficiently. This evaluation reflects overall user contentment with the system's performance, usability, and functionality.

Ethical Consideration

To ensure that the rights and welfare of all participants are upheld, the researchers will adhere to a set of ethical guidelines in conducting this study.

In order to achieve this, the following moral guidelines are followed:

Informed Consent. Participants will receive a clear and comprehensive explanation of the study's objectives, procedures, and expected outcomes. They will be informed about any potential risks or benefits associated with their participation. Ample time will be provided for them to ask questions and seek clarification before they provide their consent

Voluntary Participation. Participation in the study will be entirely voluntary. No respondent will be coerced or pressured into joining. They will have the right to withdraw at any stage of the research without any obligation to provide an explanation. Their decision to discontinue participation will not result in any negative consequences.

Non-Disruption of Operations. The study will be conducted with minimal interference in the daily business activities of trucking services. Data collection, system testing, and implementation processes will be scheduled in coordination with the participating businesses to ensure that normal operations are not disrupted. Efforts will be made to integrate research activities smoothly into existing workflows.

Anonymity. The identities of all respondents will remain anonymous to protect their privacy. No identifying information will be included in reports, presentations, or publications. Data will be coded and analyzed without linking responses to specific individuals or businesses, ensuring that participants cannot be personally identified.

Confidentiality. All collected data, including financial records and operational details of trucking businesses, will be

treated with strict confidentiality. Personal and businessrelated information will not be disclosed to unauthorized individuals. Data will be securely stored using appropriate protection measures to prevent unauthorized access, ensuring the privacy of all respondents.

Research Integrity. The researchers will uphold a high standard of honesty and accuracy in all aspects of the study. Data will be collected, analyzed, and reported objectively. Misrepresentation, fabrication, or manipulation of results will be strictly avoided. Transparency will be maintained throughout the research process to ensure credibility and reliability.

Chapter III

RESULTS AND DISCUSSION

This chapter presents the results and analysis of the data gathered from the respondents. The findings are displayed in tables, accompanied by descriptions to support and explain the findings

1. The Assessment of the Manual System

1.1 Security of Files and Records

When it comes to securing files and records, a manual system can have several limitations. Depending on manual processes to handle sensitive information can create vulnerabilities,

indicating that such methods may not offer the most dependable protection for important data.

Table 10 shows the result of the assessment made by the candidate end-users regarding the security of files and records.

Table 10

Result of the Assessment on the Security of Files and Records Using Manual System

Item Statements	Weighted Mean	Verbal Description
The manual system often leads to lost or misplaced files.	3.35	Strongly Agree
There is a lack of control over who has access to files and records.	3.35	Strongly Agree
The manual system is vulnerable to theft or unauthorized access.	3.08	Agree

Item Statements	Weighted Mean	Verbal Description
The manual system does not provide adequate backup or disaster recovery options.	3.27	Agree
The manual system is prone to errors, including data entry mistakes and transcription errors.	3.08	Agree
The manual system is difficult to update and maintain, leading to outdated or incorrect information.	3.15	Agree
The manual system does not provide a complete audit trail of who has accessed or modified records.	3.12	Agree
Item Statements	Weighted Mean	Verbal Description

The manual system does not provide the necessary confidentiality and privacy protections for sensitive information.	3.04	Agree
The manual system is time-consuming and labor-intensive to use and manage.	3.12	Agree

Grand Mean

3.17

Verbal Description

Agree

As illustrated in the table, respondents generally viewed the manual system as having issues regarding the security and management of files and records, with a computed mean of **3.17**, indicating general agreement on the presence of these problems.

According to Isaev (2025), security issues in file handling could significantly affect the long-term stability of an institution and, relying on manual methods for record management is time-consuming and labor-intensive.

The results indicate that the manual system is vulnerable, hard to maintain, and labor-intensive, which motivates the need for improvements or the implementation of a more secure and efficient system. Handling assessments manually increases the risk

of confidential information exposure, and the inefficiency of manual processes makes management more challenging and timedemanding.

1.2 Management of Data

Table 11 shows the result of the assessment made by the candidate end-users regarding the management of data using the manual system.

Item Statements	Weighted Mean	Verbal Description
The manual system results in errors in data entry.	3.19	Agree
The manual system leads to inefficient data retrieval and storage.	3.12	Agree
The manual system takes up too much time and resources.	3.27	Agree

Item Statements	Weighted Mean	Verbal Description
The manual system lacks the ability to perform complex data analysis.	3.27	Agree
The manual system is prone to data loss or corruption.	3.29	Strongly Agree
The manual system lacks data security and confidentiality.	3.38	Strongly Agree
The manual system lacks the ability to integrate with other systems.	3.35	Strongly Agree
The manual system requires manual processing and handling of	3.35	Strongly Agree
paperwork.		
The manual system lacks real-time data updates and reporting.	3.27	Agree

The manual system lacks the ability to scale and handle larger volumes of data.	3.31	Strongly Agree
---	------	----------------

Grand Mean	3.28
-------------------	------

Verbal Description	Strongly Agree
---------------------------	----------------

The results show that respondents generally agree that the manual system has several weaknesses, as reflected by the grand mean of **3.28**. The highest ratings indicate that the system consumes too much time and resources and lacks the ability to perform complex data analysis, suggesting major limitations in efficiency and capability. Users also agree that the manual system is prone to errors, data loss, security issues, and struggles with large data volumes and system integration.

These findings are supported by existing literature, which notes that manual processes often lead to delays and inconsistencies due to repetitive human input (Kim & Park, 2021). Similarly, study shows that manual data handling increases the likelihood of errors and reduces operational efficiency compared to automated systems (Fernandez & Liu, 2022).

Overall, the results and supporting studies highlight a clear need to shift from a manual system to a more automated, reliable, and secure digital process. Improving efficiency, accuracy, and data handling capacity will better support organizational operations and reduce the limitations identified by respondents.

1.3 Reliability of Output

Table 12 shows the result of the assessment made by the candidate end-users regarding the reliability of output.

Table 12

Results of the Assessment on the Reliability of Output Using Manual System

	Weighted Mean	Verbal Description
Item Statements		
The manual system produces inaccurate outputs.	3.19	Agree
The manual system is prone to errors in data entry.	3.15	Agree

The manual system is time- consuming and inefficient.	3.15	Agree
---	------	-------

The manual system requires significant effort to maintain and update.	3.12	Agree
---	------	-------

The manual system lacks consistency in output quality.	3.23	Agree
--	------	-------

The manual system is susceptible to data loss and corruption.	3.15	Agree
---	------	-------

The manual system is difficult to use for	3.27	Agree
---	------	-------

complex data
analysis.

Item Statements	Weighted Mean	Verbal Description
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The manual system is not scalable to accommodate increased workload.	3.31	Strongly Agree
The manual system is dependent on users' skills and knowledge.	3.35	Strongly Agree
The manual system lacks security measures to protect confidential data.	3.19	Agree
Grand Mean		3.21
Verbal Description		Agree

As shown in Table 12, respondents generally perceived the manual system as difficult to maintain, inconsistent, and prone to errors, with a general weighted mean of **3.21**. The highest concerns were its lack of scalability for increased workload ($\mu = 3.31$) and its dependence on users' skills and knowledge ($\mu = 3.35$), both rated **Strongly Agree**. Respondents also agreed that the manual system lacks consistent output quality ($\mu = 3.23$), produces inaccurate outputs ($\mu = 3.19$), and is susceptible to data loss and corruption ($\mu = 3.15$). Furthermore, they agreed that the system requires significant effort to maintain ($\mu = 3.12$) and is difficult to use for complex data analysis ($\mu = 3.27$).

These findings are supported by the study noting that manual processes are time-consuming, more prone to human error, and less capable of handling larger or more complex tasks (Lopez & Santos, 2022). Studies also point out that manual data handling exposes organizations to risks of data loss, inconsistencies, and workflow disruptions (Harper & Singh, 2023; Bautista, 2021).

1.4 Efficiency of Processes

Table 13 shows the result of the assessment made by the candidate end-users regarding the efficiency of processes.

Table 13

Results of the Assessment on the Efficiency of Processes Using Manual System

Item Statements	Weighted Mean	Verbal Description
The manual system is time-consuming and slows down the overall process.	3.27	Agree
Manual data entry is prone to errors and inaccuracies.	3.31	Strongly Agree
The manual system is difficult to update and maintain.	3.23	Agree

The manual system is inefficient and requires excessive use of paper and other resources.	3.12	Agree
Manual systems are less reliable than computerized systems.	3.35	Strongly Agree
Manual systems lack the automation features that are necessary for efficient data management.	3.38	Strongly Agree
Manual systems require more personnel and	3.35	Strongly Agree

resources to manage
than computerized
systems.

Manual systems lack the flexibility to adapt to changing business needs and processes.	3.27	Agree
Manual systems are more vulnerable to security breaches and data loss than computerized systems.	3.23	Agree
Manual systems lack the analytical capabilities necessary for effective data analysis and reporting.	3.15	Agree
Grand Mean	3.27	
Verbal Description	Strongly Agree	

Based on the results, the respondents find the manual system to be inefficient. The grand mean score of 3.27 shows they agree the system has major problems. Specifically, the respondents strongly believe the system lacks automation ($\mu = 3.38$) and requires more personnel and resources to

Item Statements	Weighted Mean	Verbal Description
manage ($\mu = 3.35$). They also strongly agree that manual data entry is prone to errors and inaccuracies ($\mu = 3.31$) and that manual systems are less reliable than computerized systems ($\mu = 3.35$). Additionally, they report that the system is timeconsuming ($\mu = 3.27$), difficult to update and maintain ($\mu = 3.23$), and lacks flexibility to adapt to new or changing needs ($\mu = 3.27$). They also agree that it is vulnerable to security breaches and data loss ($\mu = 3.23$) and lacks analytical capabilities for effective data analysis ($\mu = 3.15$).		

These findings match what previous researchers have found, wherein manual work is known to be slow and full of mistakes (Sharma & Singh, 2020). Also, companies that use old, manual systems often fall behind because they cannot analyze data effectively (Davenport, 2018).

1.5 Compliance with Regulation and Standards

Table 14 shows the result of the assessment made by the candidate end-users regarding the compliance with regulation and standards.

Table 14

Results of the Assessment on Compliance with Regulation and Standards Using Manual System

Compliance with regulations and standards is

difficult to maintain with a manual system.	3.15	Agree
It is challenging to ensure that all staff members are aware of and following compliance regulations and standards.	3.19	Agree
There is a risk of non-compliance due to human error in a manual system.	3.23	Agree
Manual systems lack the level of control necessary to ensure compliance with complex regulations and standards.	3.31	Strongly Agree

Item Statements	Weighted Mean	Verbal Description
Maintaining compliance with Regulations and Standards is timeconsuming in a manual system.	3.35	Strongly Agree
Compliance requirements change frequently, making it difficult to keep up to date in a manual system.	3.42	Strongly Agree
It is challenging to demonstrate compliance with regulations and standards in a manual system.	3.23	Agree
Manual systems are more prone to		

mistakes, which can
lead to compliance
issues.

3.19

Agree

The lack of
automation in a
manual system makes

3.27

Agree

it difficult to
enforce compliance
regulations and
standards.

The absence of audit
trails in a manual
system makes it
challenging to

3.31

Strongly Agree

demonstrate
compliance with
regulations and
standards.

Grand Mean

3.27

Verbal Description

Strongly Agree

Based on the results, respondents agreed that maintaining compliance with regulations and standards is difficult when using a manual system, as shown by the grand mean of **3.27**. They reported that staff often struggle to stay updated and consistent with regulatory requirements ($\mu = 3.19$), and that the manual process increases the risk of human error leading to noncompliance ($\mu = 3.23$). Respondents also noted that manual systems lack sufficient control to meet complex regulatory demands ($\mu = 3.31$) and require significant time to manage compliance tasks ($\mu = 3.35$).

The strongest concerns involved the difficulty of keeping up with frequently changing compliance requirements ($\mu = 3.42$) and demonstrating compliance due to the system's limitations ($\mu = 3.23$). Additionally, respondents agreed that manual systems are more prone to mistakes ($\mu = 3.19$), lack automation needed to enforce standards ($\mu = 3.27$), and do not provide audit trails necessary for proving compliance ($\mu = 3.31$).

These findings are consistent with studies showing that manual systems often fail to support accurate, consistent, and regulation-ready documentation, increasing the risks of compliance errors (Gibson & Richards, 2020; Kim & Park, 2021).

2. The Development of the System based on the Stages of the Agile Scrum Model.

2.1 Initiation

During the initiation phase, the researchers develop a clear vision for the study, outlining its goals and what they aim to accomplish. They also identify important stakeholders who may influence or be affected by the project, making sure their needs are considered. A team leader is assigned to guide the group throughout the project, helping maintain good communication and smooth decision-making.

The researchers then describe the main set of tasks, which represents the broad areas of work the project needs to focus on before breaking them down into smaller, manageable activities. This organized approach increases the chances of completing the system successfully, ensuring it meets user needs and is finished within the planned schedule and resources (Atlassian, 2024).

To support time management, the team also creates a Kanban board that shows the project timeline, helping them stay on track throughout the development process.

Figure 4

Gantt chart

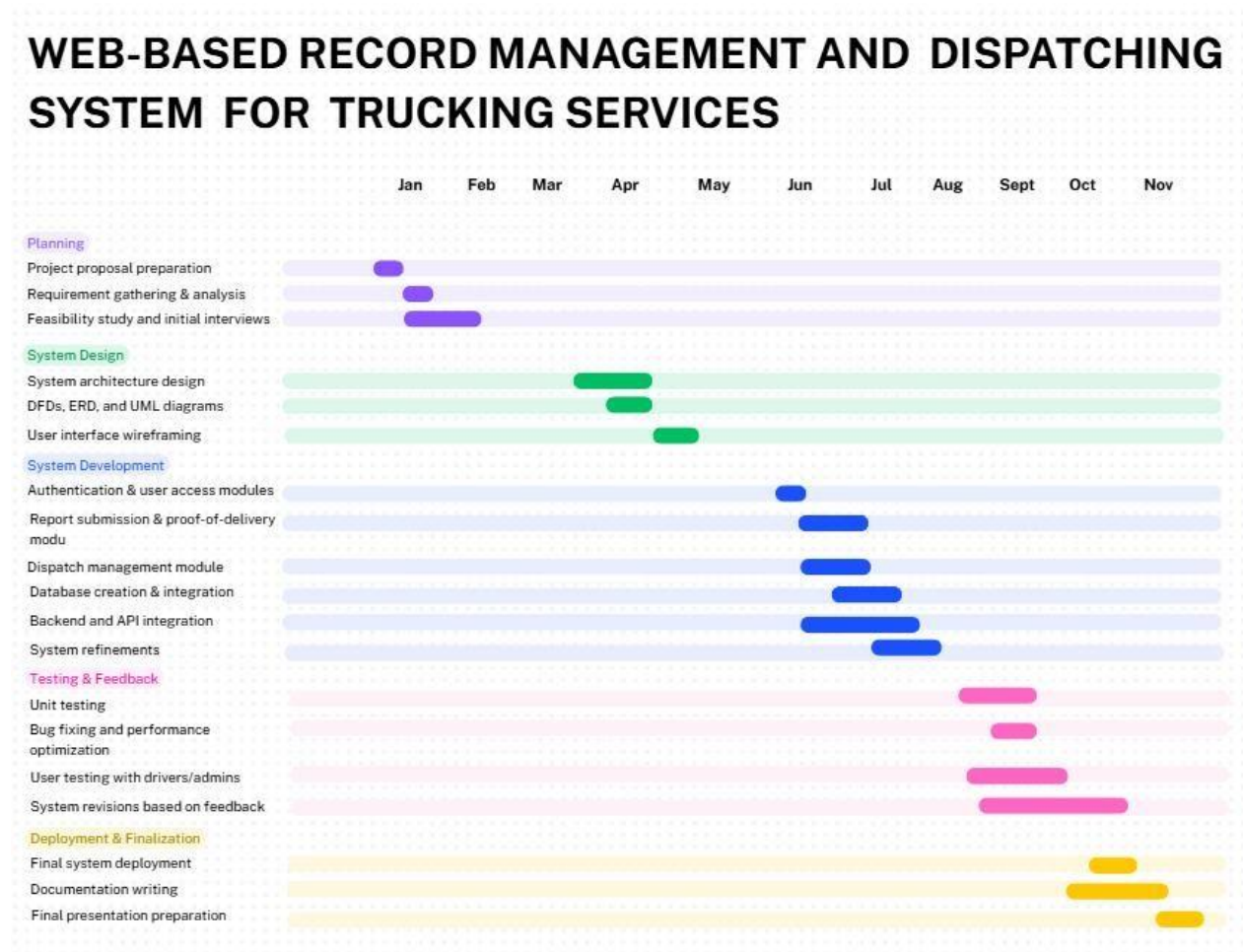


Figure4 shows the Gantt chart for the project, providing a clear timeline for each phase, including initiation, planning, implementation, reviewing, and releasing. The chart allows the researchers to track progress, manage time effectively, and stay on schedule throughout the development process. By visualizing the sequence of tasks, the Gantt chart helps the researchers identify potential delays or overlaps in work.

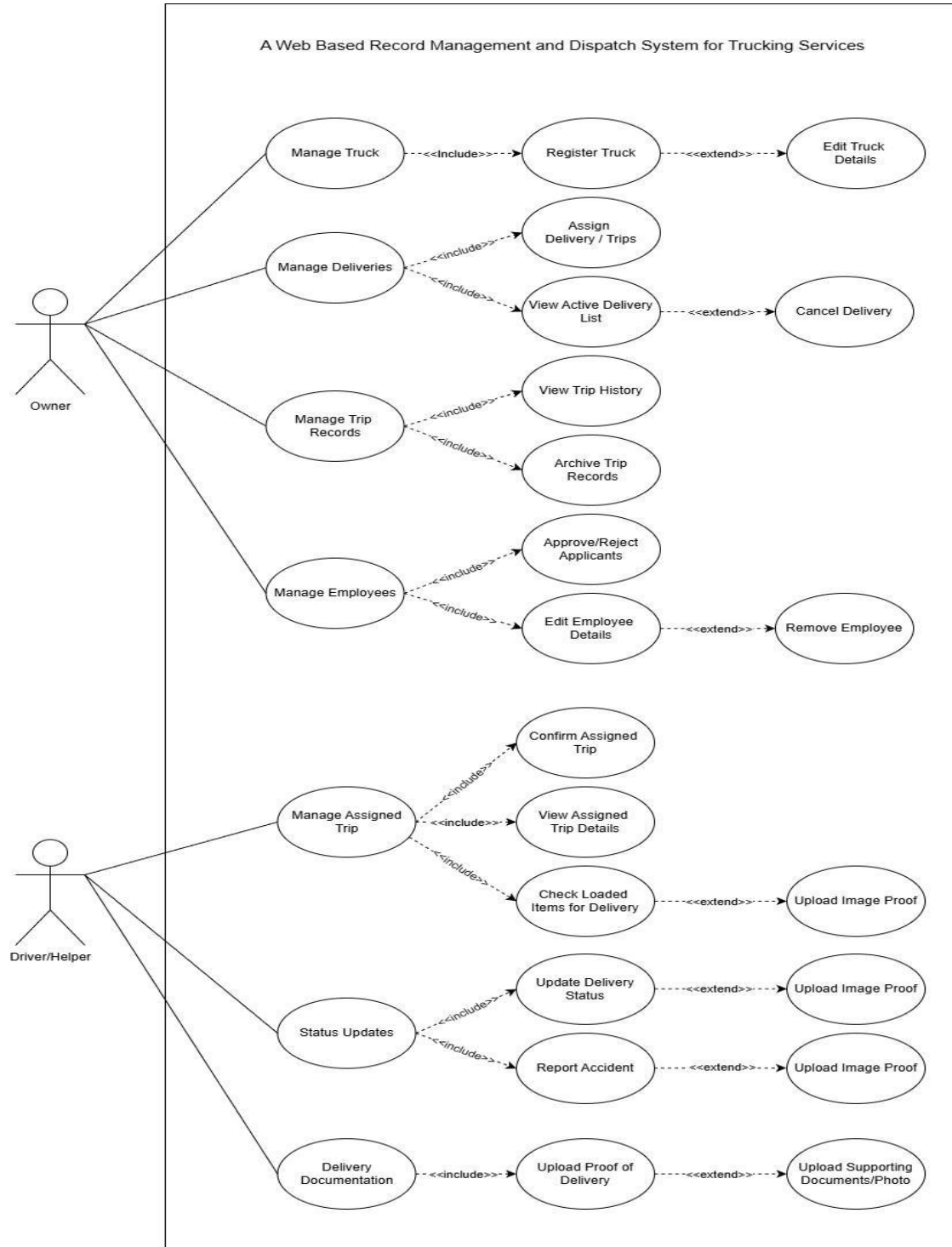
2.2 Planning

During the Planning and Estimation phase, the researchers began by creating a use case diagram to illustrate the system's functionality from the user's perspective. This was followed by designing a flowchart derived from the use case and a context diagram to show the system's interactions with external entities. Additionally, they developed Level 0 and Level 1 Data Flow Diagrams (DFDs) to represent the flow of data and the processes within the system.

Figure 5 illustrates the use case diagram for the trucking services system, highlighting the system's key functionalities and user interactions. The diagram identifies two primary roles: the Owner and the Driver/Helper, each with specific responsibilities within the system.

The Owner can manage trucks, deliveries, trip records, and employees, including tasks such as registering trucks, assigning deliveries, reviewing trip histories, and approving or editing employee details. The Driver/Helper is responsible for managing assigned trips, updating delivery status, reporting accidents, and uploading proof of delivery. Each role has distinct interactions that ensure the smooth operation of the system, while the use of include and extend relationships clarifies dependencies between tasks and optional actions, such as uploading supporting documents or editing details.

Figure 5



Database Design

The database design defines how data is organized and structured within the trucking management system. The researchers identified the necessary entities—such as deliveries, employees, trucks, invoices, and incident reports—and organized them into separate tables to ensure accuracy and efficiency. Relationships between these tables were established using primary and foreign keys, allowing the system to properly manage tasks like assigning employees, tracking deliveries, storing proofs, and generating invoices. This design ensures that all data is stored clearly, consistently, and can be accessed or updated without conflict.

Database Normalization

Database normalization was applied to arrange the data into well-structured tables while removing redundancy. By normalizing up to the Third Normal Form (3NF), the researchers ensured that each table contains only related attributes and that no unnecessary duplication exists. Information such as delivery details, employee records, and truck data was separated into their own entities to avoid inconsistencies and update problems. This normalization process keeps the database organized, easier to maintain, and more efficient for storing and retrieving records used in the trucking management system..

Figure 6

Database Normalization

Unnormalized Form

delivery_id delivery_date status created_at truck_id driver_id helper_id helper2_id client_name client_contact delivery_address delivery_items_item_id delivery_items_product_name delivery_items_quantity delivery_items_uom delivery_items_price delivery_items_total delivery_crew_employee_id delivery_crew_role proof_id submitted_at proof_status proof_truck_id proof_driver_id proof_helper_id proof_notes archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname proof_photos_photo_id proof_photos_photo_path proof_photos_uploaded_at dispatch_id dispatch_truck_id route_from route_to dispatch_date dispatch_status arrival_time completion_date estimated_arrival dispatch_archived_driver_fname dispatch_archived_driver_lname dispatch_archived_helper_fname dispatch_archived_helper_lname dispatch_archived_helper2_fname
--

dispatch_archived_helper2_fname
dispatch_archived_helper2_lname
employee_id
employee_role
first_name
middle_name
last_name
email
contact_number
address
birthdate
profile_picture
employee_created_at
employee_status
approval_status
password
assignment_id
assignment_employee_id
assignment_truck_id
assignment_role
date_assigned
assignment_status
id_counter_role
last_number
incident_id
incident_delivery_id
incident_employee_id
incident_type
other_type
incident_date
incident_time
location
latitude
longitude
description
damage_description
estimated_cost
injuries
police_report_filed
police_station
report_number
incident_created_at
archived_employee_fname
archived_employee_lname
archived_employee_role
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname

archived_helper2_fname
archived_helper2_lname
incident_files_file_id
incident_files_file_name
incident_files_file_path
incident_files_file_type
incident_files_file_size
incident_files_uploaded_at
invoice_id
invoice_number
invoice_delivery_id
customer_name
invoice_date
due_date
subtotal
shipping_fee
tax
total_amount
payment_status
remarks
invoice_created_at
invoice_items_item_id
invoice_items_description
invoice_items_quantity
invoice_items_price
pod_id
pod_delivery_id
pod_submitted_at
pod_status
pod_truck_id
pod_driver_id
pod_helper_id
pod_helper2_id
pod_notes
pod_archived_driver_fname
pod_archived_driver_lname
pod_archived_helper_fname
pod_archived_helper_lname
pod_archived_helper2_fname
pod_archived_helper2_lname
pod_photos_photo_id
pod_photos_photo_path
pod_photos_uploaded_at
registration_id
registration_role
registration_first_name
registration_middle_name
registration_last_name
registration_email
registration_contact_number
registration_address

registration_birthdate
id_number
id_image
barangay_clearance
nbi_clearance
medical_certificate
drug_test
registration_profile_picture
registration_created_at
registration_status
registration_password
report_id
report_delivery_id
report_employee_id
report_status
current_location
report_latitude
report_longitude
report_remarks
supporting_docs
report_created_at
report_archived_driver_fname
report_archived_driver_lname
report_archived_helper_fname
report_archived_helper_lname
report_archived_helper2_fname
report_archived_helper2_lname
report_archived_employee_fname
report_archived_employee_lname
report_archived_employee_role
truck_id
truck_model
truck_brand
plate_number
capacity
year_acquired
color
truck_status
classification
truck_brand_id
brand_name
truck_brand_created_at

1st Normal form

delivery_id
delivery_date
status
created_at
truck_id
driver_id
helper_id
helper2_id
client_name
client_contact
delivery_address
item_id
delivery_id
product_name
quantity
uom
price
total
photo_id
proof_id
photo_path

incident_id
delivery_id
employee_id
incident_type
other_type
incident_date
incident_time
location
latitude
longitude
description
damage_description
estimated_cost
injuries
police_report_filed
police_station
report_number
created_at
archived_employee_fname
archived_employee_lname
archived_employee_role
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname
file_id
file_name
file_path
file_type
file_size
uploaded_a

dispatch_id
truck_id
route_from
route_to
dispatch_date
status
arrival_time
completion_date
estimated_arrival
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname

employee_id
role
first_name
middle_name
last_name
email
contact_number
address
birthdate
profile_picture
created_at
employee_status
approval_status
password
assignment_id
employee_id
truck_id
role
date_assigned
status
last_number

invoice_id
invoice_number
delivery_id
customer_name
invoice_date
due_date
subtotal
shipping_fee
tax
total_amount
payment_status
remarks
created_at
item_id invoice_id
description
quantity
price

pod_id
delivery_id
submitted_at
status
truck_id
driver_id
helper_id
helper2_id
notes
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname
photo_id
pod_id photo_path
uploaded_at

registration_id

role
first_name
middle_name
last_name
email
contact_number
address
birthdate
id_number
id_image
barangay_clearance
nbi_clearance
medical_certificate
drug_test
profile_picture
created_at
status

truck_id

truck_model
truck_brand
plate_number
capacity
year_acquired
color
status
classification
id
brand_name

report_id

delivery_id
employee_id
status current_location
latitude
longitude
remarks
supporting_docs
created_at
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname
archived_employee_fname
archived_employee_lname
archived_employee_role

2nd Normal Form

DELIVERIES	INCIDENT_REPORT
delivery_id delivery_date status created_at truck_id driver_id helper_id helper2_id client_name client_contact delivery_address item_id delivery_id product_name quantity uom price total photo_id proof_id photo_path	incident_id delivery_id employee_id incident_type other_type incident_date incident_time location latitude longitude description damage_description estimated_cost injuries police_report_filed police_station report_number created_at archived_employee_fname archived_employee_lname archived_employee_role archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname archived_helper2_fname archived_helper2_lname file_id file_name file_path file_type file_size uploaded_at
DISPATCH_RECORDS	
dispatch_id truck_id route_from route_to dispatch_date status arrival_time completion_date estimated_arrival archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname archived_helper2_fname archived_helper2_lname	

EMPLOYEES
employee_id role first_name middle_name last_name email contact_number address birthdate profile_picture created_at employee_status approval_status password assignment_id truck_id role date_assigned status last_number

PROOF_OF_DELIVERY
pod_id delivery_id submitted_at status truck_id driver_id helper_id helper2_id notes archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname archived_helper2_fname archived_helper2_lname photo_id pod_id photo_path uploaded_at

INVOICES
invoice_id invoice_number delivery_id customer_name invoice_date due_date subtotal shipping_fee tax total_amount payment_status remarks created_at item_id description quantity price

REGISTRATION
registration_id role first_name middle_name last_name email contact_number address birthdate id_number id_image barangay_clearance nbi_clearance medical_certificate drug_test profile_picture created_at status

STATUS_REPORT
report_id delivery_id employee_id status current_location latitude longitude remarks supporting_docs created_at archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname archived_helper2_fname archived_helper2_lname archived_employee_fname archived_employee_lname archived_employee_role

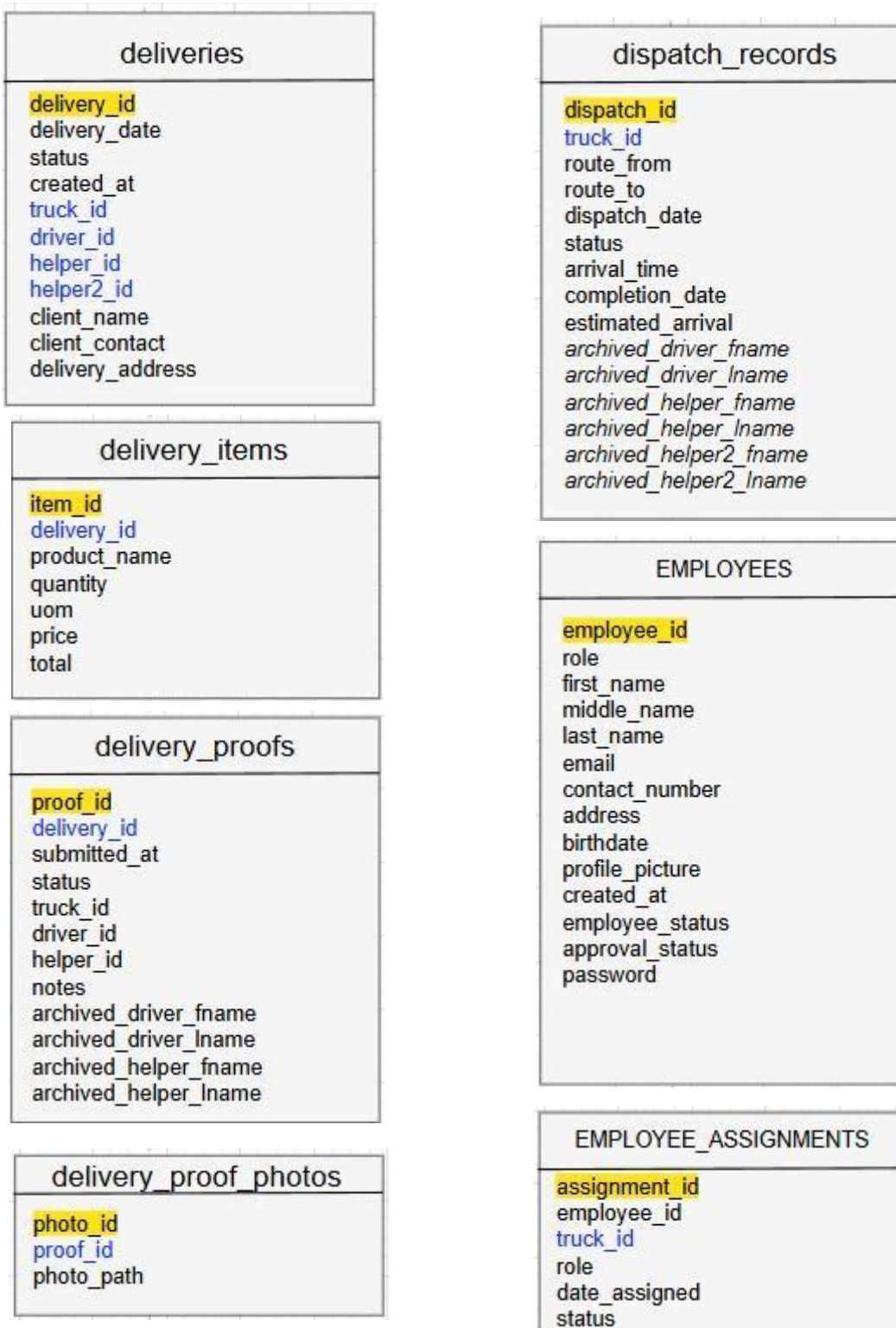
TRUCKS
truck_id truck_model truck_brand plate_number capacity year_acquired color status classification id brand_name

PROOF_OF_DELIVERY_PHOTOS
photo_id pod_id photo_path uploaded_at

TRUCKS
truck_id truck_model truck_brand plate_number capacity year_acquired color status classification

INCIDENT_REPORTS
incident_id delivery_id employee_id incident_type other_type incident_date incident_time location latitude longitude description damage_description estimated_cost injuries police_report_filed police_station report_number created_at archived_employee_fname archived_employee_lname archived_employee_role archived_driver_fname archived_driver_lname archived_helper_fname archived_helper_lname archived_helper2_fname archived_helper2_lname

3rd Normal Form



ID_COUNTER
role
last_number

INCIDENT_REPORTS
incident_id
delivery_id
employee_id
incident_type
other_type
incident_date
incident_time
location
latitude
longitude
description
damage_description
estimated_cost
injuries
police_report_filed
police_station
report_number
created_at
archived_employee_fname
archived_employee_lname
archived_employee_role
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname

INCIDENT_REPORT_FILES
file_id
incident_id
file_name
file_path
file_type
file_size
uploaded_at

TRUCK_BRANDS
id
brand_name

INVOICES
invoice_id
invoice_number
delivery_id
customer_name
invoice_date
due_date
subtotal
shipping_fee
tax
total_amount
payment_status
remarks
created_at

INVOICE_ITEMS
item_id
invoice_id
description
quantity
price

PROOF_OF_DELIVERY
pod_id
delivery_id
submitted_at
status
truck_id
driver_id
helper_id
helper2_id
notes
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname

PROOF_OF_DELIVERY_PHOTOS
photo_id
pod_id
photo_path
uploaded_at

REGISTRATIONS
registration_id
role
first_name
middle_name
last_name
email
contact_number
address
birthdate
id_number
id_image
barangay_clearance
nbi_clearance
medical_certificate
drug_test
profile_picture
created_at
status

STATUS_REPORTS
report_id
delivery_id
employee_id
status
current_location
latitude
longitude
remarks
supporting_docs
created_at
archived_driver_fname
archived_driver_lname
archived_helper_fname
archived_helper_lname
archived_helper2_fname
archived_helper2_lname
archived_employee_fname
archived_employee_lname
archived_employee_role

TRUCKS
truck_id
truck_model
truck_brand
plate_number
capacity
year_acquired
color
status
classification

Figure 6 illustrates the normalized structure of the EG Trucking database. The data used across the system—such as deliveries, trucks, employees, invoices, and incident reports—are organized into separate, dedicated tables. This separation reduces data duplication by ensuring that each piece of information is stored only once, making the entire database easier to manage and maintain.

Through proper normalization, EG Trucking is able to handle large amounts of operational data more efficiently. It ensures consistency when updating records, prevents errors caused by redundant information, and improves the reliability of functions such as tracking deliveries, managing assignments, generating invoices, and monitoring truck activity. As a result, the system becomes more stable, accurate, and effective in supporting the company's daily operations.

Entity-Relationship Diagram

The Entity-Relationship (ER) Diagram represents the conceptual and logical structure of the database used in the system. It illustrates the different entities, their corresponding attributes, and the relationships that connect them. By using an ER diagram, the flow of data and the interactions between components become clearer, serving as a guide for designing and implementing an organized and efficient database.

Figure 7

Entity-Relationship Diagram

Field Name	Size	Data Type	Description
delivery_id	11	INT	Primary key, autoincrementing delivery identifier
delivery_date		DATE	Scheduled date for delivery
status		ENUM	Current status of the delivery
created_at		TIMESTAMP	Timestamp when delivery record was created
truck_id	20	VARCHAR	Foreign key referencing assigned truck
driver_id	20	VARCHAR	Foreign key referencing assigned driver
helper_id	20	VARCHAR	Foreign key referencing first helper
helper2_id	20	VARCHAR	Foreign key referencing second helper Name of the
client_name	255	VARCHAR	

Table: 15

client/customer			
Field Name	Size	Data Type	Description
client_contact	20	VARCHAR	number of the Contact client
delivery_address		TEXT	Complete delivery address

Table: 16

Data Dictionary of delivery_crew

Field Name	Size	Data Type	Description
id	11	INT	autocrew Primary key, assignment incrementing identifier
delivery_id	11	INT	Foreign key referencing delivery
employee_id	20	VARCHAR	Foreign key referencing employee
role		ENUM	Role of employee in this delivery

Table: 17

Data Dictionary of delivery_items

Field Name	Size	Data Type	Description
item_id	11	INT	Primary key, auto-incrementing item
Field Name	Size	Data Type	Description
			identifier
delivery_id	11	INT	Foreign key referencing delivery
product_name	255	VARCHAR	Name of the product/item
quantity	50	VARCHAR	Quantity of items delivered
uom	10	VARCHAR	Unit of measurement
price	10,2	DECIMAL	Price per unit
total	10,2	DECIMAL	Total price (quantity × price)

Table: 18

Data Dictionary of delivery_proofs

Field Name	Size	Data Type	Description
proof_id	11	INT	Primary key

delivery_id	11	INT	Foreign key referencing delivery
submitted_at		TIMESTAMP	Timestamp when proof was submitted
status		ENUM	Approval status of the proof

Field Name	Size	Data Type	Description
truck_id	20	VARCHAR	Truck used for delivery
driver_id	20	VARCHAR	Driver who submitted proof
helper_id	20	VARCHAR	Helper who assisted
notes		TEXT	1 notes or Addition comments
archived_driver_fname	100	VARCHAR	Archived driver first name
archived_driver_lname	100	VARCHAR	Archived driver last name
archived_helper_fname	100	VARCHAR	Archived helper first name
archived_helper_lname	100	VARCHAR	Archived helper last name

Table: 19

Data Dictionary of delivery_proof_photos

Field Name	Size	Data Type	Description
photo_id	11	INT	Primary key
proof_id	11	INT	Foreign key referencing delivery proof

Field Name	Size	Data Type	Description
photo_path	255	VARCHAR	File path to uploaded photo
uploaded_at		TIMESTAMP	Timestamp when photo was uploaded

Table: 20

Data Dictionary of dispatch_records

Field Name	Size	Data Type	Description
dispatch_id	11	INT	Primary key
truck_id	20	VARCHAR	Foreign key referencing dispatched truck
route_from	255	VARCHAR	Starting location of route
route_to	255	VARCHAR	Destination location

dispatch_date		DATETIME	Date and time of dispatch
status		ENUM	Current dispatch status
arrival_time		DATETIME	Actual arrival time
completion_date		DATETIME	Date and time of completion
estimated_arrival		DATETIME	Estimated arrival time
archived_driver_fname	100	VARCHAR	Archived driver first

Field Name	Size	Data Type	Description
			name
archived_driver_lname	100	VARCHAR	Archived driver last name
archived_helper_fname	100	VARCHAR	Archived helper first name
archived_helper_lname	100	VARCHAR	Archived helper last name
archived_helper2_fname	100	VARCHAR	Archived helper second first name
archived_helper2_lname	100	VARCHAR	Archived helper second last name

Table: 21

Data Dictionary of employees

Field Name	Size	Data Type	Description
employee_id	20	VARCHAR	Primary key, employee identifier
role		ENUM	Employee role in system
first_name	50	VARCHAR	Employee's first name
middle_name	50	VARCHAR	Employee's middle name

Field Name	Size	Data Type	Description
last_name	50	VARCHAR	Employee's last name
email	100	VARCHAR	Employee's email
contact_number	20	VARCHAR	Employee's contact number
address		TEXT	Employee's complete address
birthdate		DATE	Employee's birthdate
profile_picture	255	VARCHAR	Path to profile picture
created_at		TIMESTAMP	Timestamp when employee record was created
employee_status		ENUM	Current employment status
approval_status		ENUM	Approval status for system access
password	255	VARCHAR	Encrypted password for login

Table: 22

Data Dictionary of employee_assignments

Field Name	Size	Data Type	Description
------------	------	-----------	-------------

assignment_id 11 INT Primary key employee_id 20 VARCHAR Foreign
 key referencing employee truck_id 20 VARCHAR Foreign key
 referencing assigned

Field Name	Size	Data Type	Description
			truck
role		ENUM	Role in this assignment
date_assigned		DATE	Date when assignment was made
status		ENUM	Current assignment status

Table: 23

Data Dictionary of id_counter

Field Name	Size	Data Type	Description
role		ENUM	Primary key, employee role type
last_number	11	INT	Last used number for ID generation

Table: 24

Data Dictionary of incident_reports

Field Name	Size	Data Type	Description
incident_id	11	INT	Primary key
			Foreign key

delivery_id	11	INT	referencing related delivery
employee_id	20	VARCHAR	Foreign key

Field Name	Size	Data Type	Description
incident_type		ENUM	referencing reporting employee Type of incident
other_type	255	VARCHAR	Description if incident type is 'Other'
incident_date		DATE	Date when incident occurred
incident_time		TIME	Time when incident occurred
location	255	VARCHAR	Location where incident occurred
latitude	10,8	DECIMAL	GPS latitude coordinate
longitude	11,8	DECIMAL	GPS longitude coordinate
description		TEXT	Detailed description of incident
damage_description		TEXT	Description of damages incurred
estimated_cost	10,2	DECIMAL	Estimated cost of damages
injuries	255	VARCHAR	Description of any injuries

Field Name	Size	Data Type	Description
police_report_filed		ENUM	Whether police report was filed
police_station	255	VARCHAR	Name of police station
report_number	100	VARCHAR	Police report number
created_at		TIMESTAMP	Timestamp when report was created
archived_employee_fname	100	VARCHAR	Archived employee first name
archived_employee_lname	100	VARCHAR	Archived employee last name
archived_employee_role	50	VARCHAR	Archived employee role
archived_driver_fname	100	VARCHAR	Archived driver first name
archived_driver_lname	100	VARCHAR	Archived driver last name
archived_helper_fname	100	VARCHAR	Archived helper first name
archived_helper_lname	100	VARCHAR	Archived helper last name
archived_helper2_fname	100	VARCHAR	Archived second helper first name Archived second helper

Field Name	Size	Data Type	Description
archived_helper2_lname			100 VARCHAR last name

Table: 25

Data Dictionary of incident_report_files

Field Name	Size	Data Type	Description
file_id	11	INT	Primary key
incident_id	11	INT	Foreign key referencing incident report
file_name	255	VARCHAR	Original name of uploaded file
file_path	500	VARCHAR	Path to stored file
file_type	100	VARCHAR	MIME type
file_size	11	INT	File size in bytes
uploaded_at		TIMESTAMP	Timestamp when file was uploaded

Table: 26

Data Dictionary of invoices

Field Name	Size	Data Type	Description
invoice_id	11	INT	Primary key

Field Name	Size	Data Type	Description
invoice_number	50	VARCHAR	Unique invoice number

Field Name	Size	Data Type	Description
delivery_id	11	INT	Foreign key referencing related delivery
customer_name	255	VARCHAR	Name of customer
invoice_date		DATE	Invoice date
due_date		DATE	Payment due date
subtotal	10,2	DECIMAL	Subtotal amount
shipping_fee	10,2	DECIMAL	Shipping fee
tax	10,2	DECIMAL	Tax amount
total_amount	10,2	DECIMAL	Total amount due
payment_status		ENUM	Current payment status
remarks		TEXT	Additional notes
created_at		TIMESTAMP	Timestamp when invoice was created

Data Dictionary of invoice_items

Field Name	Size	Data Type	Description
item_id	11	INT	Primary key

invoice_id	11	INT	Foreign key
description	255	VARCHAR	Description of invoiced item

Table: 28

Field Name	Size	Data Type	Description
quantity	11	INT	Quantity of items
price	10,2	DECIMAL	Price per unit

Table: 27

Field Name	Size	Data Type	Description
pod_id	11	INT	Primary key
delivery_id	11	INT	Foreign key
submitted_at		TIMESTAMP	Timestamp when POD was submitted
status		ENUM	Approval status
truck_id	20	VARCHAR	Truck used
driver_id	20	VARCHAR	Driver
helper_id	20	VARCHAR	First helper
helper2_id	20	VARCHAR	Second helper
notes		TEXT	Additional notes

archived_driver_fname	100	VARCHAR	Archived driver first name
archived_driver_lname	100	VARCHAR	Archived driver last name

Data Dictionary of proof_of_delivery

			name
Field Name	Size	Data Type	Description
archived_helper_fname	100	VARCHAR	Archived helper first name

archived_helper_lname	100	VARCHAR	Archived helper last name
archived_helper2_fname	100	VARCHAR	Archived second helper first name
archived_helper2_lname	100	VARCHAR	Archived second helper last name

Table: 29

Data Dictionary of proof_of_delivery_photos

Field Name	Size	Data Type	Description
photo_id	11	INT	Primary key
pod_id	11	INT	Foreign key
photo_path	255	VARCHAR	Path to uploaded photo
uploaded_at		TIMESTAMP	Timestamp when photo was uploaded

Table: 30

Data Dictionary of registrations

Field Name	Size	Data Type	Description
registration_id	20	VARCHAR	Primary key
role		ENUM	Role applied for
first_name	50	VARCHAR	First name

middle_name	50	VARCHAR	Middle name
last_name	50	VARCHAR	Last name
email	100	VARCHAR	Email
contact_number	20	VARCHAR	Contact number
address		TEXT	Complete address
birthdate		DATE	Birthdate
id_number	50	VARCHAR	Government ID number
id_image	255	VARCHAR	Path to ID image
barangay_clearance	255	VARCHAR	Path to barangay clearance
nbi_clearance	255	VARCHAR	Path to NBI clearance
medical_certificate	255	VARCHAR	Path to medical certificate
drug_test	255	VARCHAR	Path to drug test result
profile_picture	255	VARCHAR	Path to profile picture
created_at		TIMESTAMP	Record timestamp

Field Name	Size	Data Type	Description
status		ENUM	Registration approval status
password	255	VARCHAR	Encrypted password

Table: 31

Data Dictionary of status_reports

Field Name	Size	Data Type	Description
report_id	11	INT	Primary key
delivery_id	11	INT	Foreign key
employee_id	20	VARCHAR	Reporting employee
status		ENUM	Current delivery status
current_location	255	VARCHAR	Current location
latitude	10,8	DECIMAL	GPS latitude
longitude	11,8	DECIMAL	GPS longitude
remarks		TEXT	Remarks
supporting_docs	255	VARCHAR	Path to documents
created_at		TIMESTAMP	Timestamp created
			Archived driver

archived_driver_fname 100 VARCHAR

first name

Field Name	Size	Data Type	Description
------------	------	-----------	-------------

archived_driver_lname	100	VARCHAR	Archived driver last name
-----------------------	-----	---------	---------------------------

archived_helper_fname	100	VARCHAR	Archived helper first name
-----------------------	-----	---------	----------------------------

archived_helper_lname	100	VARCHAR	Archived helper last name
-----------------------	-----	---------	---------------------------

archived_helper2_fname	100	VARCHAR	Archived second helper first name
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archived_helper2_lname	100	VARCHAR	Archived second helper last name
------------------------	-----	---------	----------------------------------

archived_employee_fname	100	VARCHAR	Archived employee first name
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archived_employee_lname	100	VARCHAR	Archived employee last name
-------------------------	-----	---------	-----------------------------

archived_employee_role	50	VARCHAR	Archived employee role
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Table: 32

Data Dictionary of trucks

Field Name	Size	Data Type	Description
truck_id	20	VARCHAR	Primary key
truck_model	100	VARCHAR	Model of the truck
Field Name	Size	Data Type	Description
truck_brand	100	VARCHAR	Brand
plate_number	20	VARCHAR	License plate number
capacity	10,2	DECIMAL	Load capacity
year_acquired		DATE	Year acquired
color	50	VARCHAR	Color
status		ENUM	Operational status
classification	20	VARCHAR	Truck classification

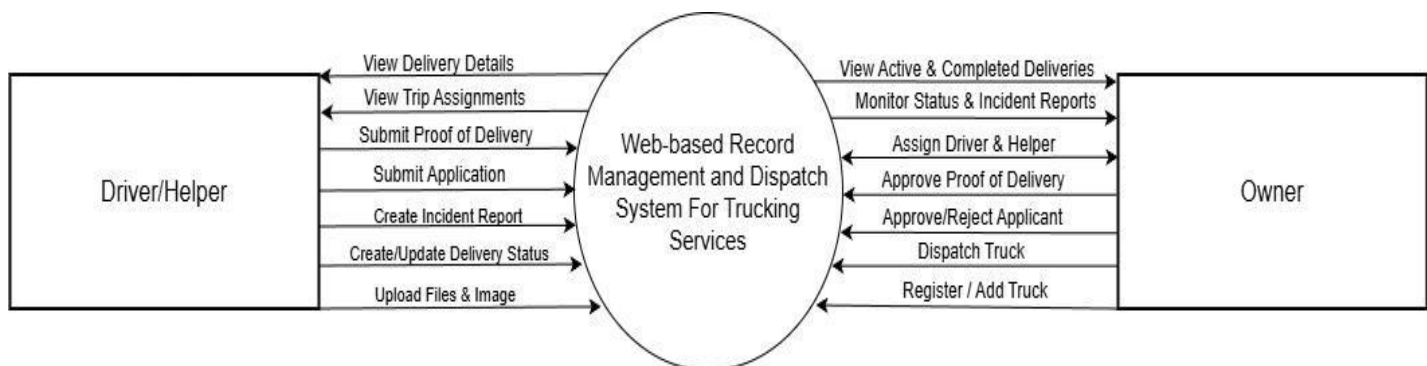
The data dictionary is a collection of metadata used within the trucking management system. It serves as an important reference for understanding the structure of the database, helping developers and users manage the system efficiently, and ensuring that all data related to deliveries, trucks, drivers, and operations remain accurate, consistent, and reliable.

2.3 Development

The development phase focuses on implementing the system's functional and non-functional requirements. During this stage, a Data Flow Diagram (DFD) is used to illustrate how data moves throughout the system. The DFD identifies the system's major components, including external entities, processes, data flows, and data stores, providing a clear visualization of how information is processed and managed within the system.

Context Diagram

A Context Diagram is the highest level of a Data Flow Diagram (DFD) that presents a simplified, big-picture view of how data moves through a system. It represents the entire system as a single process and shows how it interacts with external entities, without detailing the internal processes.

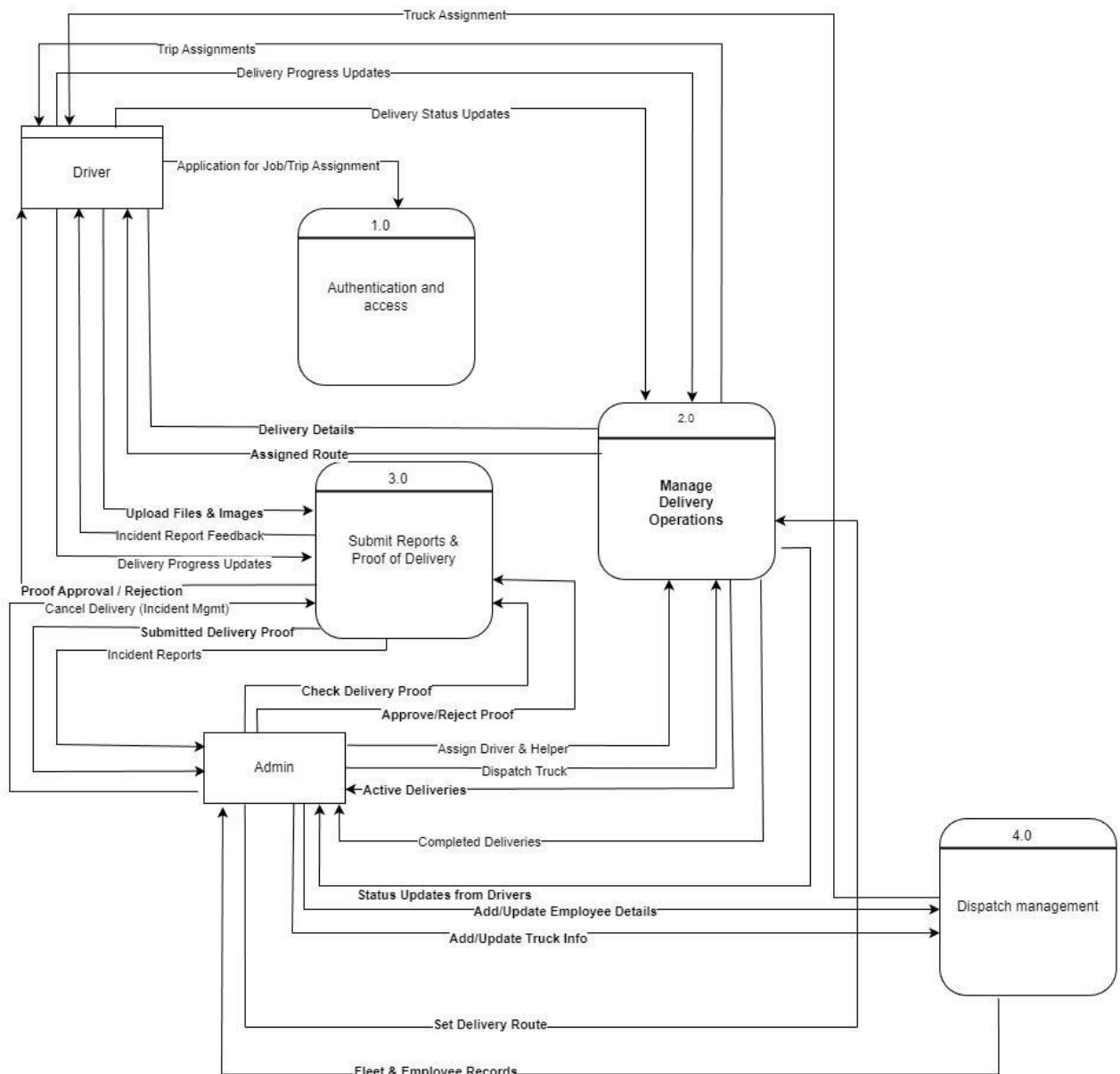


Data Flow Diagram - Level 0

A level 0 Data Flow Diagram shows the main processes in a system and their interactions with the external entities and data stores.

Figure 8

Data Flow Diagram - Level 0



DFD LEVEL0

Figure 8 shows the overall data flow within the Eg Trucking Management System. The DFD Level 0 highlights four main processes: Authentication and Access (1.0), Manage Delivery Operations (2.0), Submit Reports & Proof of Delivery (3.0), and Dispatch Management (4.0).

Drivers interact with the system by requesting trip assignments, updating delivery status, uploading proof of delivery, and submitting incident reports. Administrators manage truck and route assignments, monitor active and completed deliveries, check submitted proofs, and update fleet and employee records.

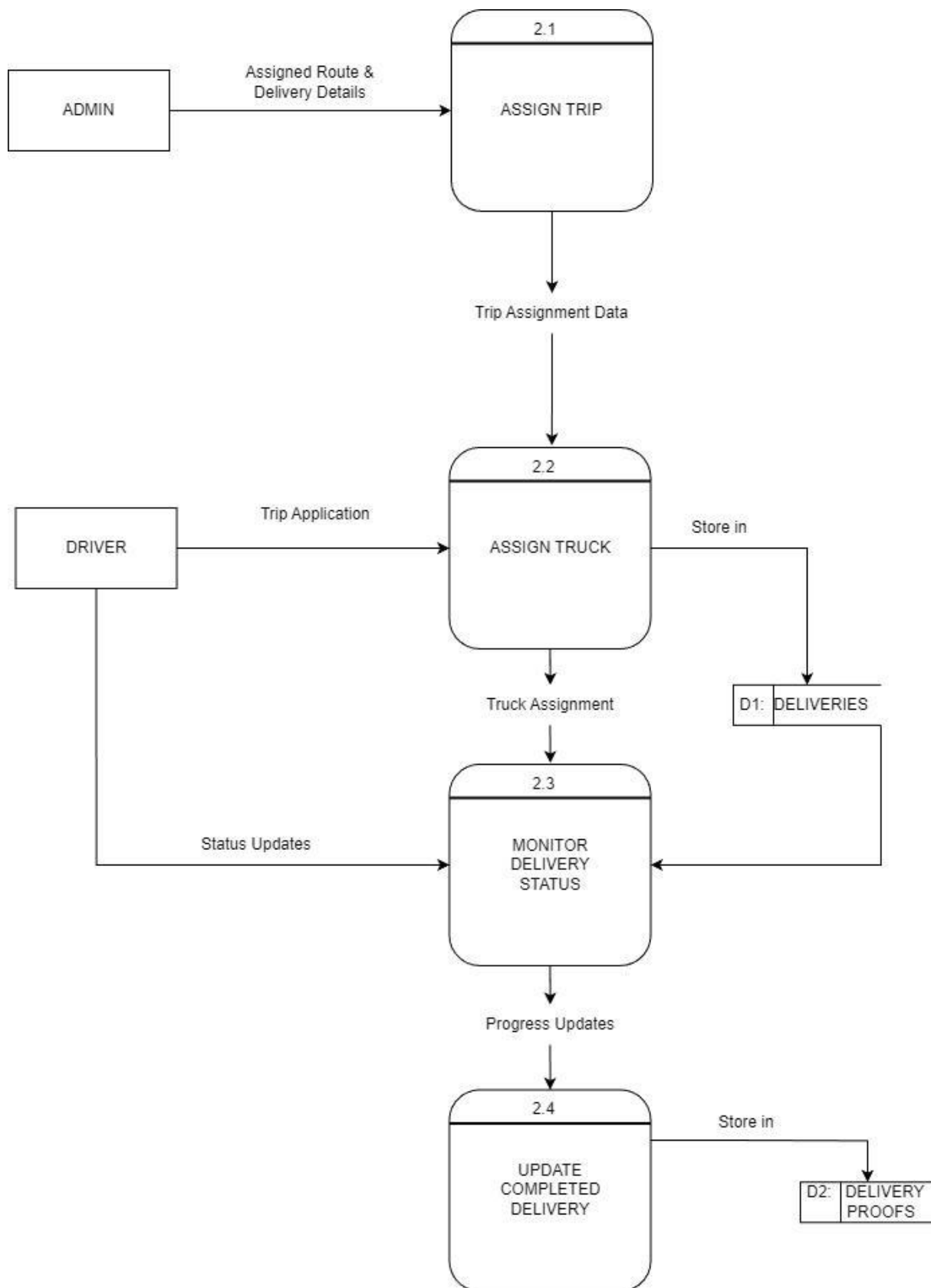
The diagram illustrates how information moves between drivers, admins, and the system processes, forming a clear structure of how Eg Trucking manages its delivery operations. This data flow provides the foundation for ensuring organized, accurate, and efficient trucking activities.

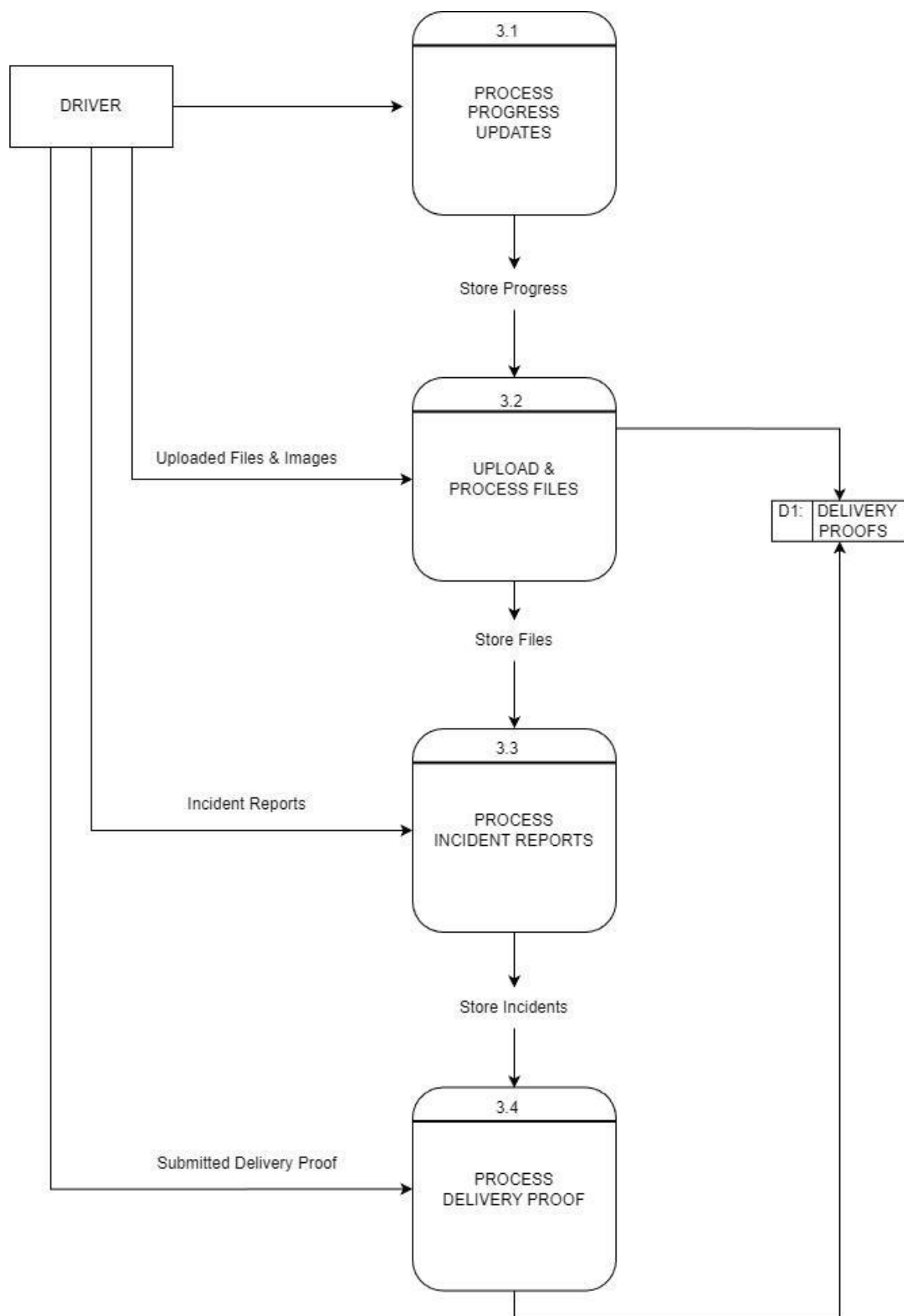
Data Flow Diagram - Level 1

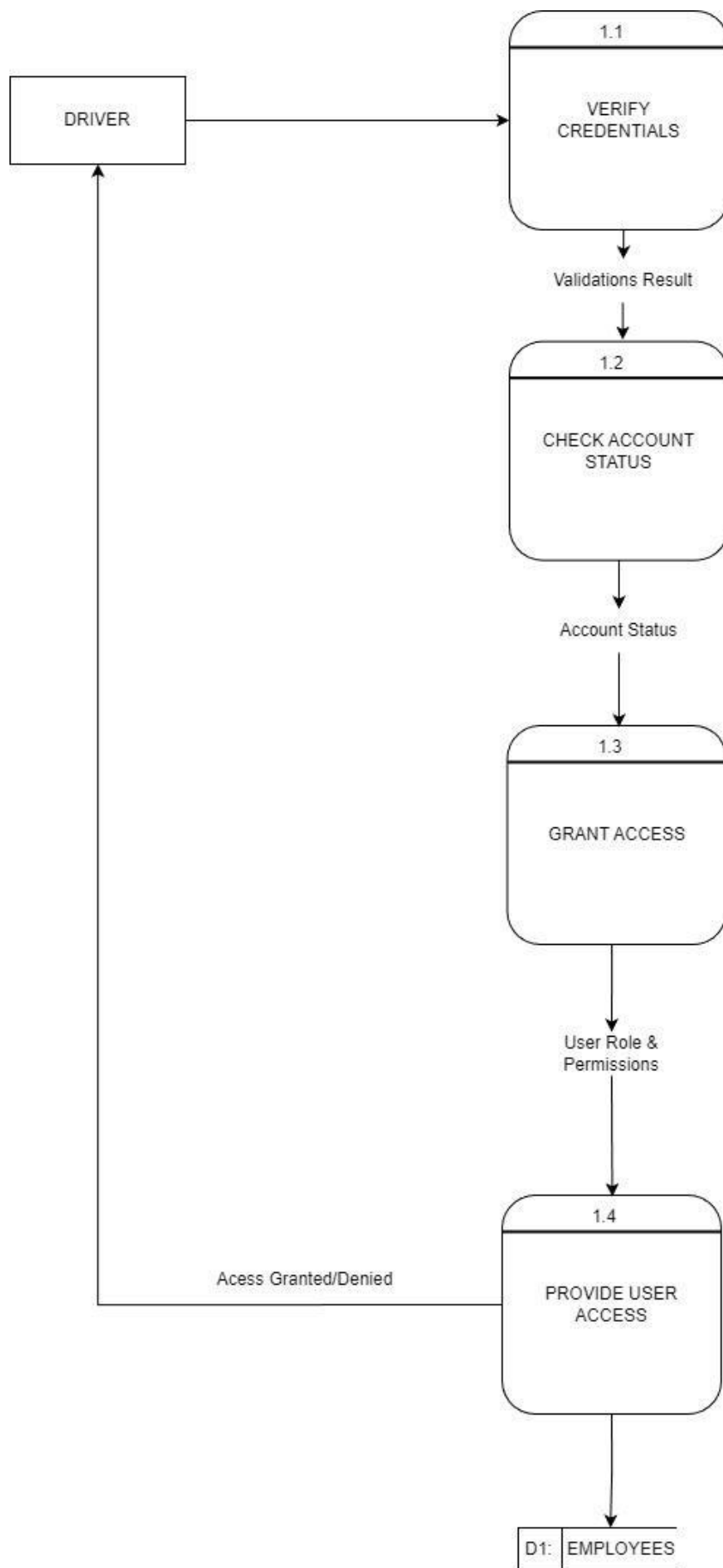
Level 1 of the Data Flow Diagram shows the breakdown of the main processes into sub-processes. It illustrates the components in a more detailed manner than a level 0 DFD. In this diagram, the system's subprocesses are revealed.

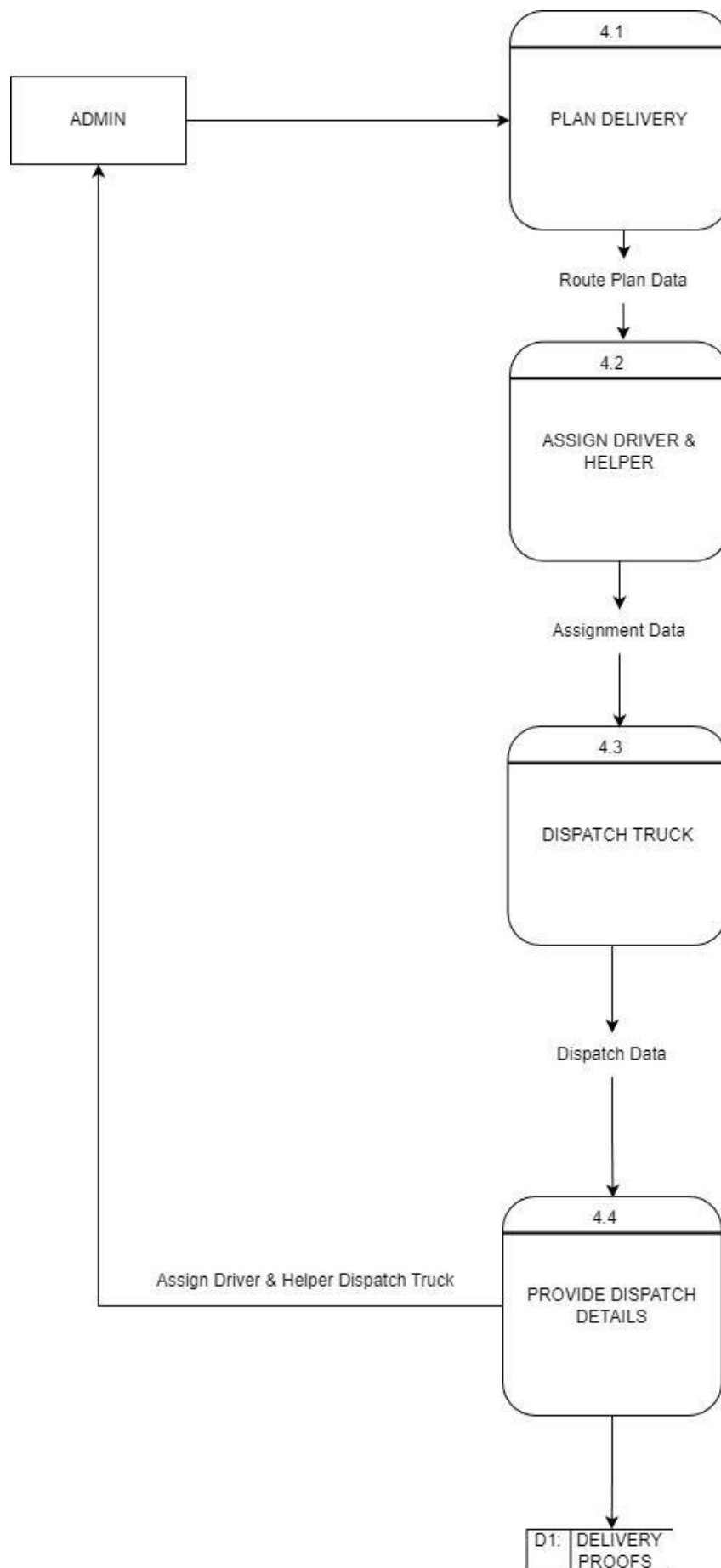
Figure 9

Data Flow Diagram - Level 1









3. The Assessment of the System's Technical Qualities by elected IT Experts in Compliance with Software Characteristics Guidelines

The following tables show the results of evaluation by the elected IT Experts concerning the technical aspects of the system.

3.1 Functional Suitability

Table 33 shows the result of the assessment made by the elected IT Experts regarding the functional suitability of the system.

Table 33

Results of the Assessment Made by elected IT Experts on the Systems' Functional Suitability

Item Statement	Weighted Mean	Verbal Description
Functional Completeness: The set of functions covers all the specified tasks and intended users' objectives.	3.70	Highly Suitable
Functional Correctness: The system provides accurate results when used by intended users.	3.70	Highly Suitable
Functional Appropriateness: The functions help users accomplish specified tasks and objectives efficiently.	3.60	Highly Suitable

Grand Mean	3.67
Verbal Description	Highly Suitable

As presented in Table 33, the assessment made by the respondents indicates that the system is highly suitable, as reflected in the overall weighted mean of 3.67. The results show that the system's set of functions adequately covers all specified tasks and user objectives under Functional Completeness ($\mu = 3.70$, Verbal Description = Highly Suitable). Respondents also confirmed that the system provides accurate results when used by its intended users under Functional Correctness ($\mu = 3.70$, Verbal Description = Highly Suitable). In addition, they agreed that the system's functions effectively support users in accomplishing specified tasks and goals under Functional Appropriateness ($\mu = 3.60$, Verbal Description = Highly Suitable).

According to Ariyani et al. (2021), functional suitability refers to the extent to which a system delivers functions that satisfy user requirements under specified conditions. This quality characteristic is evaluated through three sub-dimensions: functional completeness, which assesses how fully a system covers specified tasks; functional correctness, which determines the accuracy of system outputs; and functional appropriateness, which measures how effectively a system assists users in achieving their objectives (Silva et al., 2022). These evaluations help developers and organizations optimize system performance, enhance user satisfaction, and maintain competitiveness in rapidly evolving technological environments (Hasanah et al., 2020).

Overall, the results imply that the respondents find the system highly suitable in terms of functional performance. It sufficiently addresses all specified tasks, consistently delivers

accurate results, and supports users in efficiently completing their designated objectives.

3.2 Performance Efficiency

Table 34 shows the result of the assessment made by the elected IT Experts regarding the performance efficiency of the system.

Table 34

Results of the Assessment Made by elected IT Experts on the Systems' Performance Efficiency

Item Statement	Weighted Mean	Verbal Description
Time Behavior: The system responds quickly when performing its functions (e.g., loading records, assigning trips, or viewing dispatch data).	3.50	Excellent
Resource Utilization: The system efficiently uses resources (e.g., server, database, and bandwidth) even when accessed by multiple users.	3.80	Excellent

Item Statement	Weighted Mean	Verbal Description
Capacity: Degree to which the system can handle multiple users,	3.80	Excellent
transactions, and concurrent activities without performance degradation		
Grand Mean		3.70

Verbal Description	Excellent
--------------------	-----------

As shown in Table 34, the respondents rated the system's Performance Efficiency with an overall grand mean of 3.70, interpreted as Excellent. The system demonstrated strong performance across all three sub-characteristics.

For Time Behavior, the system received a weighted mean of 3.50, indicating that it responds quickly when performing tasks such as loading records and viewing dispatch data. Resource Utilization obtained a higher mean of 3.80, showing that the system uses server, database, and bandwidth resources efficiently even with multiple users. Likewise, Capacity was rated 3.80, reflecting that the system can handle several users, transactions, and simultaneous activities without performance issues.

Performance efficiency, according to Ariyani et al. (2021), evaluates how well a system performs under given resource conditions. It includes time behavior, resource utilization, and capacity—metrics that help determine whether the system remains fast, stable, and scalable (Silva et al., 2022; Hasanah et al., 2020).

Overall, the findings indicate that the system delivers excellent performance, maintaining quick responses, efficient

resource usage, and reliable performance even under higher workloads.

3.3 Compatibility

Table 35 shows the result of the assessment made by the elected IT Experts regarding the compatibility of the system.

Table 35

Results of the Assessment Made by elected IT Experts on the Systems' Compatibility

Item Statement	Weighted Mean	Verbal Description
Co-existence: The system performs its required functions efficiently while sharing a common environment (e.g., database, or web browser) with other applications without causing conflicts or errors.	3.50	Highly Compatible hosting server, database, or web browser)
Interoperability: The system can effectively exchange and use information with other systems, such as company Highly Compatible databases, vehicle tracking systems, or external tools for documentation and communication.	3.40	
Grand Mean	3.45	
Verbal Description	Highly Compatible	

Table 35 illustrates that the respondents rated the system as highly compatible, as reflected in the computed grand mean of 3.55. Specifically, the respondents indicated that the system is capable of performing its essential functions efficiently within a shared

Item	Statement	Weighted Mean	Verbal Description
	environment such as a hosting server, database, or web browser—without causing conflicts or errors with other applications ($\mu = 3.50$, Verbal Description = Highly Compatible).		

Furthermore, respondents acknowledged that the system demonstrates strong interoperability, particularly in its ability to effectively exchange and utilize information with other systems. This includes integration with company databases, vehicle tracking tools, or external applications for documentation and communication ($\mu = 3.40$, Verbal Description = Highly Compatible).

Compatibility serves as an essential indicator of system reliability, encompassing the concepts of co-existence and interoperability. Co-existence refers to the system's ability to share resources with other applications in a common environment without disruption, while interoperability ensures that the system can communicate and exchange data meaningfully with other platforms (Sherkar et al., 2021).

The findings suggest that the system exhibits a high degree of compatibility in both aspects. It performs smoothly in shared environments and collaborates effectively with external systems, making it suitable for operations that require integrated processes and coordinated information exchange.

3.4 Usability

Table 36 shows the result of the assessment made by the elected IT Experts regarding the usability of the system.

Table 36

Results of the Assessment Made by elected IT Experts on the Systems' Usability.

Item Statement	Weighted Mean	Verbal Description
Appropriateness		
Recognizability: Users can easily recognize that the system is designed to meet their needs in managing trucking operations, such as recording data, assigning trips, and tracking dispatch activities.	3.60	Excellent
Learnability: Users can quickly learn how to use		

Item Statement	Weighted Mean	Verbal Description
the system and its main functions without requiring extensive training or technical guidance.	3.50	Excellent
Operability: The system is easy to operate, allowing users to control and navigate through different modules (e.g., records, dispatch, trip details) smoothly and efficiently.	3.60	Excellent
User Error Protection: The system minimizes user errors (e.g., incorrect data entry, accidental deletions) and provides clear alerts or feedback messages when mistakes occur.	3.30	Excellent
User Engagement: The interface is visually organized and encourages users to interact with	3.80	Excellent

Item Statement	Weighted Mean	Verbal Description

the system, making their work experience more efficient and engaging.

Inclusivity: The system can be effectively used by different types of users (e.g., dispatchers, drivers, and administrators) regardless of their technical skill levels.

Self-Descriptiveness: The system clearly indicates how each feature works and what actions users can take, reducing the need for additional explanations or manuals.	3.50	Excellent
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	3.60	Excellent
Grand Mean		3.61

Verbal Description	Excellent
---------------------------	------------------

Table 36 illustrates that the respondents rated the system as excellent in usability, as reflected in the computed grand mean of 3.61. Specifically, the respondents indicated that users can easily recognize that the system is designed to meet their needs in managing trucking operations, such as recording data, assigning trips, and tracking dispatch activities ($\mu = 3.60$, Verbal Description = Excellent). Furthermore, respondents

acknowledged that the interface is visually organized and encourages users to interact with the system, making their work experience more efficient and engaging ($\mu = 3.80$, Verbal Description = Excellent).

Usability serves as a critical factor in system adoption and user satisfaction, encompassing aspects such as learnability, operability, and user engagement. According to Nielsen (2020), systems with high usability reduce training time, minimize errors, and increase productivity. The high ratings in learnability ($\mu = 3.50$) and operability ($\mu = 3.60$) indicate that users can quickly understand and navigate the system without extensive training, while the strong performance in user engagement ($\mu = 3.80$) suggests an intuitive and motivating user interface (Wilson et al., 2021).

The findings suggest that the system exhibits outstanding usability across multiple dimensions. It effectively supports users with varying technical skill levels, provides clear guidance on feature usage, and minimizes operational errors, making it highly suitable for diverse users in trucking operations, including dispatchers, drivers, and administrators (Zhang & Li, 2022).

3.6 Security

Table 37 shows the result of the assessment made by the elected IT Experts regarding the security of the system.

Table37

Results of the Assessment made by elected IT Experts on the Systems' Security

Item Statement	Weighted Mean	Verbal Description
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Security Confidentiality:
The system ensures that

sensitive information (such as driver details, trip records, and user accounts) is accessible only to authorized personnel and protected from unauthorized disclosure.	3.70	Strong Security
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Integrity: The system maintains the accuracy and consistency of records by preventing unauthorized modifications, deletions, or corruption of data related to dispatch operations and trip management.	3.60	Strong Security
--	------	-----------------

Item Statement	Weighted Mean	Verbal Description
-----------------------	----------------------	---------------------------

Non-repudiation: The system records and keeps detailed activity logs and transaction histories

to ensure that all actions (such as trip assignments, approvals, and updates) are verifiable and cannot be denied by the users who performed them.	3.40	Strong Security
--	------	-----------------

Accountability: The system tracks and identifies user activities, allowing administrators to monitor who performed specific actions within the system for transparency and security purposes.	3.70	Strong Security
---	------	-----------------

Authenticity: The system verifies user identities and ensures that only Security legitimate and verified accounts can access or perform operations within the system.	3.80	Strong
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Grand Mean	3.68
Verbal Description	Strong Security

Table 37 illustrates that the respondents rated the system as having strong security, as reflected in the computed grand mean of

3.68. Specifically, the respondents indicated that the system ensures sensitive information—such as driver details, trip records, and user accounts—is accessible only to authorized personnel and protected from unauthorized disclosure ($\mu = 3.70$, Verbal Description = Strong Security). Furthermore, respondents acknowledged that the system maintains the accuracy and consistency of records by preventing unauthorized modifications, deletions, or corruption of data related to dispatch operations and trip management ($\mu = 3.60$, Verbal Description = Strong Security).

Security serves as an essential indicator of system reliability, encompassing the concepts of confidentiality, integrity, nonrepudiation, accountability, and authenticity. According to Alzahrani et al. (2022), confidentiality ensures that sensitive data remains accessible only to authorized users, while integrity guarantees that information remains accurate and unaltered. Nonrepudiation ensures that all user actions are logged and verifiable, accountability enables tracking of user activities, and authenticity confirms that only legitimate users can access the system (Kumar & Singh, 2021).

The findings suggest that the system exhibits a high degree of security across multiple dimensions. It effectively protects sensitive data, maintains record accuracy, and ensures user accountability, making it suitable for operations that require secure and reliable information management in trucking services (Chen & Wang, 2020).

4. The Assessment of the System's Quality of Use by candidate end-users in compliance with Software Characteristics Guidelines

4.1 Ease of Use

The table 38 presents the results of the assessment made by the candidate end-users regarding the system's ease of use.

Results of the Assessment Made by Candidate End-Users on the System's Ease of Use

Item Statement	Weighted Mean	Description
The system is easy to use without requiring much effort.	3.12	Agree
Navigating through the system is straightforward.	3.15	Agree
The system provides clear and understandable instructions.	3.12	Agree
I can complete tasks without the need for additional help or guidance.	2.96	Agree
appealing	3.00	Agree
Grand Mean		3.07

Verbal Description
The interface design is simple and visually

Agree

Table 38 illustrates that the respondents rated the system as having good usability in terms of ease of use, as reflected in the computed grand mean of 3.07. Specifically, the respondents indicated that the system is easy to use without requiring much effort ($\mu = 3.12$, Verbal Description = Agree). Furthermore, respondents acknowledged that navigating through the system is straightforward ($\mu = 3.15$, Verbal Description = Agree), and the interface design is simple and visually appealing ($\mu = 3.00$, Verbal Description = Agree).

Ease of use serves as a fundamental aspect of system usability, directly impacting user adoption and productivity. According to Davis (1989) in the Technology Acceptance Model, perceived ease of use significantly influences users' willingness to adopt new systems. The consistent "Agree" ratings across all ease-of-use dimensions indicate that users find the system accessible and manageable for daily operations. While there is room for improvement to achieve "Strongly Agree" ratings, the current performance demonstrates satisfactory usability that supports efficient task completion in trucking operations (Venkatesh et al., 2003).

The findings suggest that the system provides a functional and approachable user experience. The positive ratings in navigation and interface design show that users can effectively interact with the system, though enhancements in instructional clarity and

independent task completion could further improve the overall user experience for dispatchers, drivers, and administrative staff.

4.2 Efficiency

Table 39 shows the result of the assessment made by the candidate end-users regarding the Efficiency of the system.

Results of the Assessment Made by candidate end-users on the Systems' Efficiency

Item Statement I	Weighted Mean	Description
can accomplish tasks quickly when using the system.	2.96	Agree
The system responds promptly to my actions.	3.08	Agree
I do not need to perform unnecessary steps to complete a task.	3.08	Agree
complete my work efficiently.	3.08	Agree
The time it takes to load or process		

actions is acceptable.	3.04	Agree	The system helps me
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Grand Mean

3.05

Verbal Description

Agree

Table 39 illustrates that the respondents rated the system as efficient in supporting task completion, as reflected in the computed grand mean of 3.05. Specifically, the respondents agreed that the system responds promptly to their actions ($\mu = 3.08$, Verbal Description = Agree), does not require unnecessary steps to complete tasks ($\mu = 3.08$, Verbal Description = Agree), and helps them complete work efficiently ($\mu = 3.08$, Verbal Description = Agree).

Efficiency is a critical component of system usability, particularly in operational environments like trucking services where time-sensitive tasks are common. According to Hornbæk (2006), efficiency measures how quickly users can perform tasks once they have learned the system, directly impacting workplace productivity. The consistent "Agree" ratings across all efficiency dimensions indicate that users find the system responsive and streamlined for dispatch and record management operations. While the system demonstrates satisfactory performance, the ratings suggest potential for optimization in task execution speed and processing time to achieve higher efficiency levels (Nielsen, 2012).

The findings indicate that the system provides reliable performance for daily operational needs. The positive ratings in system responsiveness and workflow efficiency show that the system

effectively supports users in managing trucking operations, though enhancements in processing speed could further improve user satisfaction among dispatchers and administrative staff.

4.2 Learnability

The table 40 presents the results of the assessment made by the candidate end-users regarding the system's learnability.

Results of the Assessment Made by Candidate End-Users on the System's Learnability

Item Statement	Weighted Mean	Description
Learning to use the system was quick and easy.	2.96	Agree
I can recall how to use the system after a period of not using it.	3.00	Agree
The system provides sufficient guidance for first-time users.	3.08	Agree
I was able to learn how to use the system without much frustration.	3.04	Agree
The icons, labels, and buttons make it easy to understand the system's functions.	3.12	Agree

Grand Mean3.04 Verbal Description Agree

Table 40 illustrates that the respondents rated the system as having good learnability, as reflected in the computed grand mean of 3.04. Specifically, the respondents agreed that the system provides sufficient guidance for first-time users ($\mu = 3.08$, Verbal Description = Agree) and that the icons, labels, and buttons make it easy to understand the system's functions ($\mu = 3.12$, Verbal Description = Agree).

Learnability is a fundamental aspect of system usability, particularly for applications used in dynamic environments like trucking services where users may have varying levels of technical expertise. According to Nielsen (2012), learnability measures how quickly new users can accomplish basic tasks the first time they encounter the design. The consistent "Agree" ratings across all learnability dimensions indicate that users can effectively understand and operate the system with minimal training. The positive assessment of visual elements such as icons and labels suggests an intuitive interface design that supports user comprehension, though there is potential for enhancement in reducing the learning curve for new users (Shneiderman et al., 2018).

4.2 User Satisfaction

The table 41 presents the results of the assessment made by the candidate end-users regarding the system's user satisfaction.

Results of the Assessment Made by Candidate End-Users on the System's User Satisfaction

Item Statement	Weighted Mean	Description
I feel comfortable using this system.	3.08	Agree
I am satisfied with the overall experience of using the system.	2.88	Agree
The system meets my expectations for a well-designed application. I would recommend this system to others who need similar	2.88	Agree
I would use this system regularly if needed.	2.96	Agree
Grand Mean		2.98
Verbal Description		Agree

Table 42 shows that the respondents rated the system as having a good level of user satisfaction, as reflected in the computed grand mean of 2.98. The users agreed that they feel comfortable using the system ($\mu = 3.08$, Verbal Description = Agree) and expressed satisfaction with their overall experience ($\mu = 2.88$, Verbal Description = Agree). They also agreed that the system meets their expectations for a well-designed application ($\mu = 2.88$, Verbal Description = Agree) and that they would recommend it to others who may need a similar system ($\mu = 2.96$, Verbal Description =

Agree). Additionally, the respondents indicated that they would use the system regularly when needed ($\mu = 3.08$, Verbal Description = Agree).

User satisfaction is a vital component in evaluating system quality, as it reflects the user's overall perception of the system's effectiveness, usability, and ability to meet their needs. According to ISO 25010, satisfaction is influenced by factors such as comfort, perceived usefulness, and the likelihood of continued usage. The consistent "Agree" ratings across all indicators suggest that users perceive the system as reliable, user-friendly, and beneficial to their tasks, particularly in operational environments such as trucking services. While the results are positive, the slightly lower means in some areas indicate opportunities for further refinement to enhance user delight and exceed expectations.

Chapter IV

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the study's findings, the conclusions drawn from the analysis, and the recommendations for future enhancement and implementation of the Web-Based Record Management and Dispatching System for Trucking Services.

Summary

This capstone project, titled "**Web-Based Record Management and Dispatching System for Trucking Services**," was developed to address the significant operational inefficiencies in the Philippine trucking industry, particularly the reliance on outdated manual processes for invoicing, record-keeping, and dispatch coordination. The study employed a descriptive developmental research design, utilizing the Agile

Scrum methodology to ensure iterative development, continuous feedback, and alignment with user needs.

The system was designed to semi-automate critical business processes for a selected trucking service in Meycauayan, Bulacan. Its key features include semi-automated invoicing and payroll, expense tracking, dispatch coordination with a visual dashboard, route optimization, proof of delivery submission, and secure financial record management. The development process followed systematic phases: assessment of the manual system, planning and design, development and initial testing, and final testing and deployment.

An evaluation of the existing manual system revealed significant challenges, with respondents strongly agreeing on its vulnerabilities in security, data management, reliability, process efficiency, and regulatory compliance. In contrast, the developed system was assessed highly by elected IT experts, receiving excellent ratings in **Functional Suitability, Performance Efficiency, Compatibility, Usability, and Security**. Furthermore, candidate end-users evaluated the system positively, agreeing that it was easy to use, efficient, learnable, and satisfactory for their daily operational needs.

The project successfully delivered a robust, web-based platform that enhances accuracy, reduces redundant data entry, optimizes resource allocation, and supports the digital transformation of local trucking businesses.

Conclusions

Based on the comprehensive development and evaluation of the system, the following conclusions were drawn:

- **Successful Development Using Agile Methodology:** The Web-Based Record Management and Dispatching System was successfully developed and implemented using the Agile Scrum framework. This iterative approach allowed for continuous refinement based on stakeholder feedback, resulting in a system that effectively addresses the specific pain points of the trucking company.
- **Effective Addressal of Manual System Deficiencies:** The system directly mitigates the critical shortcomings of the manual process, including poor security of files, inefficient data management, error-prone outputs, slow operational processes, and difficulties in maintaining regulatory compliance.
- **High Technical Quality Assurance:** The assessment by IT experts confirmed that the system exhibits high levels of functional suitability, performance efficiency, compatibility, usability, and security, meeting established software quality standards (ISO 25010).
- **Positive End-User Adoption:** The evaluation by candidate endusers (owners, drivers, helpers) demonstrated that the system is accessible, efficient to use, easy to learn, and satisfactory. This indicates a high potential for successful adoption and integration into daily trucking operations.
- **Tangible Operational Benefits:** The system provides significant benefits to stakeholders by streamlining financial management, simplifying dispatch coordination, improving record accuracy, and offering a centralized platform for operational data, thereby supporting better decision-making and long-term business sustainability.

Recommendations

To further enhance the system's efficacy, scalability, and impact, the following recommendations are proposed:

- **Integration of Advanced Payment Gateways:** Implement integrated payment gateways (e.g., GCash, Maya, bank transfers) to facilitate faster and more secure online transactions for clients, moving beyond manual payment tracking and improving cash flow.
- **Implementation of Advanced Data Analytics:** Incorporate a dedicated analytics and reporting dashboard with data visualization tools. This would allow business owners to gain deeper insights into key performance indicators (KPIs) such as fuel efficiency, driver performance, vehicle utilization, and profit margins per trip.
- **Enhancement of Route Optimization with Real-Time Data:** Upgrade the route optimization feature to integrate real-time traffic data and weather updates. This would provide drivers with dynamic routing suggestions to avoid delays, further reducing fuel consumption and improving delivery times.
- **Strengthening of System Scalability and Security:** To prepare for business growth and increased data load, migrate the system to a cloud-based infrastructure. Concurrently, conduct regular security audits and enhance cybersecurity measures, such as implementing multi-factor authentication (MFA), to protect against evolving threats.
- **Expansion of System Scope for Future Research:** Future researchers can build upon this system by exploring the integration of emerging technologies such as Artificial Intelligence (AI) for predictive maintenance of vehicles,

Internet of Things (IoT) sensors for real-time cargo monitoring, or Blockchain for creating tamper-proof supply chain records.



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